

The fundamental relation between firm market and accounting values

Synopsis

The research problem

Studies of the value relevance of accounting numbers typically use additive-linear models to estimate the relation between market values and accounting numbers. These models produce coefficient estimates that are hard to interpret, exhibit high volatility, and vary widely from study to study, leaving no closure on how the market-accounting relation should be measured.

Motivation

We aim to resolve this lack of closure by showing that the way market and accounting variables are measured, together with the invariance properties we require of estimated response coefficients under changes of measurement scale, implies that the market-accounting relation should take the form of a multiplicative power law, with response coefficients being estimated as elasticities using log-linear, not additive-linear, models.

Target population

Accounting and finance researchers conducting value relevance and fundamental analysis research, and other users of market-to-book and accounting-ratio models of firm value.

Adopted methodology

Using Compustat data for U.S. firms over 1972–2025, we first replicate the coefficient patterns produced by traditional additive-linear models of the form $M_t = \nu + b_i \sum_i A_{i,t}$. We then estimate the log-linear models implied by the multiplicative form $M_t = \nu \prod_i |A_i|_t^{\beta_i}$ and compare the resulting elasticity estimates to the additive-linear coefficients.

Findings

Book value, earnings, and dividends generally explain over 80 percent of long-run market value and nearly all of long-run returns when estimated in log-linear form. The market-to-book ratio

can be decomposed as a function of fundamental accounting-book ratios, a decomposition that additive-linear models cannot achieve with precision. Partitioning on the sign of income shows elasticities on profits are positive and on the absolute value of losses are lower or negative, and the elasticity on common dividends is virtually always positive, consistent with a signalling role for dividends in capital market asset pricing.

Keywords: *Fundamentals, Log-Linear Models, Value Relevance.*

Data Availability: Data used are available from public sources identified in the study.

JEL codes: *C10, C46, M41.*

1 Introduction

Numerous papers have examined the empirical relation between stock market values and accounting numbers to assess the ‘value relevance’ of the latter for the former. Value relevance studies are common, see for example [Holthausen and Watts \(2001\)](#); [Core, Guay and Van Buskirk \(2003\)](#); [Oswald \(2008\)](#); [Song, Thomas and Yi \(2010\)](#); [Balachandran and Mohanram \(2011\)](#); [Yu \(2013\)](#) and [Barth, Li and McClure \(2017\)](#). Despite this, they are not uncontroversial, with reservations being expressed about the validity and reliability of estimated accounting response coefficients as in, for instance, [Barth, Beaver and Landsman \(2001\)](#) and [Holthausen and Watts \(2001\)](#).

An important problem contributing to lack of closure regarding value relevance studies is the functional form of the models employed to estimate the market value response coefficients of accounting variables. These models produce coefficient estimates that are hard to interpret, exhibit great volatility, and change from study to study. For example, [Barth \(1994\)](#) reports a value of 0.79 for the book value coefficient from a regression model of per-share market values on accounting variables. Using similar models, [Francis and Schipper \(1999\)](#) report an average value of 25 (6.70) for the book value (earnings) coefficient; and [Zeng, Lee and Zhang \(2016\)](#) report a average value of 0.0421 (5.9509) for the book value (earnings) coefficient.

This paper shows that this variation in coefficient estimates is the result of the adoption of inappropriate additive-linear functional forms for the statistical models used to estimate response coefficients. We demonstrate that, because of the way we measure market and accounting variables and the invariance properties that we demand of estimated response coefficients when we change our units of measurement, the market-accounting relation is best modeled by using the form of a multiplicative power law. This implies that log-linear models, not additive-linear models, should be used to estimate market-accounting response coefficients as elasticities, not as the simple linear per-unit of change measures usually estimated in accounting research.¹

We do not present a new or alternative theory of firm value that may or may not have superior predictive ability. Instead, we present something more general. We show that any

models of firm value that relate market to accounting values using data of the type described here are constrained to be most appropriately represented by a particular functional form and that this implies the relevant response coefficient to estimate in such models is an elasticity.

Using data from U.S. firms over the period 1972 to 2025, we replicate the coefficient patterns we observe in traditional value relevance research, based on additive-linear models of the form $M_t = \nu + b_i \sum_i A_{i,t}$. We then demonstrate that log-linear models, resulting from transforming the multiplicative form $M_t = \nu \prod_i |A_i|_t^{\beta_i}$ by logs, produce superior coefficient estimates.

We show that the additive-linear form of the market-accounting relation is inconsistent with the way we measure the relevant data. The distributional form of accounting numbers also does not remotely approach the normal distribution. Log-linear cross-section estimates are easier to interpret and less volatile than their additive-linear counterparts. Multiplicative power law theory provides an accurate way to decompose the behaviour of the market to book ratio into accounting ratios. Taking the logs of absolute values of the accounting variables gives a logically consistent and empirically useful way of using information in ‘negative’ accounting variables such as losses and dividends in the fundamental analysis of firm market value. One reason why researches may hesitate to use log-linear models is that they are believed to restrict modelling to the use of variables with positive values only. We show this concern is not justified.

The evidence presented in this paper relies on two assumptions, one relating to the type of scales used to measure the variables of the market-accounting relation, the other to the way accounting variables are generated. The first assumption, **(A1)**, is that it makes sense to talk about market and accounting variables as being, say, ‘twice as much’ as another variable and that market-accounting ratios are invariant to changes of scales used to measure those variables.² The second assumption, **(A2)**, is that accounting variables are generated by an exponential growth process. From these two simple assumptions all the results in this paper follow.

Our paper contributes to a better understanding of the fundamental relationship between firm value and accounting variables and thus to value relevance and other related research areas

such as the usefulness of market and accounting ratios in financial statement analysis. These have been an important theme in the accounting literature for many years, e.g. [Easton \(1985\)](#); [Penman \(1992\)](#), [Ohlson \(1995\)](#), and, more recently, [Hirshleifer, Hsu and Li \(2013\)](#) and [Onali, Ginesti and Vasilakis \(2017\)](#), and in the discourse following [Lo and Lys \(2000\)](#); [Holthausen and Watts \(2001\)](#) and [Barth et al. \(2001\)](#). Application of the approach to estimation advocated in this paper provides more accurate and transparent models of the market-accounting relation that are easy to replicate. This is also important because replication of results continues to be a serious problem in existing approaches to research in all the areas mentioned.

2 The basis for the log-linear model

We deduce from our assumptions that long-run firm market value at time t , M_t , is related to its book value of equity, B_t by the functional forms:

$$M_t = \kappa |B_t|^\beta = \nu \prod_i |A_i|_t^{\beta_i} \quad (1)$$

where A_i , $i = 0, 1, \dots$ are lognormally distributed accounting variables that reconcile the opening book value of equity B_{t-1} with its closing book value B_t . $A_{0,t}$ is used to denote B_{t-1} . κ is a constant scale factor relating M to $|B|^\beta$, and ν is a constant scale factor relating M to the product of the $|A_i|^{\beta_i}$.³ We define the reconciliation between B_t and B_{t-1} as follows:

$$B_t = B_{t-1} + E_t - D_t - C_t + O_t \quad (2)$$

where E_t is net income, D_t is common dividends, C_t is preferred dividends, or coupons, and O_t is other items, a balancing amount with a value that ensures the arithmetic identity in (2).⁴

Thus:

$$\{A_0..A_4\} = \{B_{t-1}, E_t, D_t, C_t, O_t\} \quad (3)$$

$|B|$ and $|A_i|$ are absolute values of the unscaled accounting variables B and A_i respectively. β and β_i are elasticities showing the response of market returns in the form $\frac{\delta M}{M}$ to the proportional

change in the accounting variables, $\frac{\delta|B|}{|B|}$ and $\frac{\delta|A_i|}{|A_i|}$, respectively. See [Table 1](#) for the variable definitions.

[Table 1 about here]

Assumption **A1** supports the mathematical form of (1): market values and accounting variables are measured on ratio scales and the market-accounting relationship is continuous and invariant to changes in these scales⁵. Assumption **A2** supports the use of ordinary least squares to estimate the parameters in (1): accounting values are generated by proportional growth and decay processes.

We document empirical evidence supporting the two assumptions and their implications. Logarithms of the data transform (1) to a log-linear model that is estimated by OLS using Compustat data from 1972 to 2025. The estimates show that the main fundamental accounting variables of book value, earnings, and dividends generally explain over 80 percent of long-run market value, gauged by regression R^2 s and nearly all of long-run returns gauged by their estimated elasticities. In large samples, the market to book ratio can be decomposed as a function of fundamental accounting-book ratios. This decomposition cannot be achieved with any precision by additive-linear models. Partitioning the data on the sign of the income number shows that the elasticities on profits are positive and on the absolute values of losses are lower or negative. This demonstrates that no information is lost in taking the logs of absolute values of the accounting variables to estimate elasticities. The partitioning also shows the impact of common dividends on returns is virtually always positive and explains about 20 percent of returns in years in which dividends are paid. The positive sign on the elasticity for common dividends is strong evidence of their signalling role in capital market asset pricing.

3 Literature review

The form of the third term in (1) is that of the Cobb-Douglas production function ([Cobb and Douglas, 1928](#)). Although a commonplace model in economics, e.g. [Heijdra \(2009\)](#); [Mas-Colell, Whinston and Green \(1995\)](#), the importance of (1) as a logical law relating variables measured on a ratio scale does not appear to have been recognized in the literatures in accounting and finance. The origins of the understanding of this law are to be found in the literature on

psycho-physical scaling and the work of [Stevens and Galanter \(1957\)](#) and more formally in the theory of representational measurement, e.g. [Krantz, Luce, Suppes and Tversky \(1971\)](#). The proof of the power law principle embodied in the first assumption of the theory sketched in the next section is based on [Luce \(1959\)](#) as adapted by [Roberts \(1979\)](#). The argument depends on the analysis of functional equations by Cauchy in the early nineteenth century⁶.

The predominant source of models for the market-accounting relation in the accounting literature is based on an additive-linear theory of market value, estimated using untransformed data (e.g. [Barth, Beaver and Landsman \(1998\)](#); [Song et al. \(2010\)](#); [Balachandran and Mohanram \(2011\)](#); [Barth et al. \(2017\)](#)). Containing market and accounting variables as in (1) its form is

$$M_t = v + \sum_i b_i A_{i,t} \quad (4)$$

such that the parameters are assumed to be reliable representatives of the constant and slope coefficients in a linear relation. It has been recognized for many years that empirical findings make it difficult to interpret the parameters, notably the response coefficients, that this model produces (e.g. [Barth and Kallapur \(1996\)](#); [Easton \(1998\)](#); [Barth and Clinch \(2009\)](#)). A commonly adopted practice to trying to relate the coefficients of (4) to market values is to ‘deflate’ them by opening market values ([Christie, 1987](#)). However, that practice produces many outliers, unexpected magnitudes for estimates of the response coefficients and additionally complicates statistical inferences in other ways ([Easton and Sommers, 2003](#); [Barth and Clinch, 2009](#); [Powell, Shi, Smith and Whaley, 2009](#)).

Surprisingly little research on fundamentals discusses the merits of a general transformation of all the variables by logs, although the similarities of economic processes with those that generate the lognormal distribution have been discussed in detail ([Samuelson, 1973](#); [Johnson, Kotz and Balakrishnan, 1994](#)).

The model in (1) is the long-run solution to an underlying dynamic error-correction model. It gives theoretical long-run estimates of the market-accounting parameters. This is demonstrated in a more general econometrics context by [Pesaran and Smith \(1995\)](#) and [Madsen \(2005\)](#).

The multiplicative theory in (1) also offers closure for the discussion on ‘levels’ versus ‘changes’ or ‘return’ regressions (Ali and Zarowin, 1992; Easton and Harris, 1991). If additive-linear model (4) is a valid ‘levels’ description of the market-accounting relation, estimates of its response coefficients should measure the change in market value with respect to a per-unit change in the accounting variable. However, problems have been encountered in trying to reconcile estimates of such per-unit response coefficients to their theoretically expected values, e.g. Penman and Yehuda (2009). This deficiency in the additive-linear model is used as a point of comparison in testing our hypotheses in Section 6.

We show below that the problems with the additive-linear model is that it is an algebraically mis-specified form of the market accounting relation, which renders the interpretation the coefficient estimates practically impossible. Two reasons may explain why multiplicative theories of fundamentals that underlie the log-linear model have not yet been more widely developed in the accounting literature. One is that it requires taking the logs of the absolute values of variables, and it is believed that this leads to a loss of information contained in the signs of the variables being modeled (e.g. see Dechow, Hutton and Sloan (1999)). Another reason is that it is assumed, because book value disaggregates into additive components, the function reflecting the market-accounting relation must also take an additive form (Ohlson, 1995; Penman, 1998).

The next sections explain why both reasons are incorrect. We show that a scale-invariant, market-accounting relation should take the form of a multiplicative power law that can be estimated as a log-linear model in the magnitude of accounting variables and, further, that it satisfies the statistical conditions for valid and reliable estimation by OLS.

4 Theory and method

4.1 Theory

We stated in the introductory sections a point that it is important to emphasise in placing the subject matter contained in this paper in context within the asset valuation literature. Our theory is one about the correct way to measure the response coefficients in the relation between market and accounting values. It is not, for instance, a theory about why the coefficients on the

book value of equity may be greater or less than those on earnings or dividends. Such theories, however, can only be tested with any confidence if measures of the accounting response coefficients are valid and reliable. The theory presented here, therefore, is a precursor to the scientific testing of any substantive theory that attempts to explain or predict the systematic effect of accounting on market values. The import of the theory explained in this section is that, unless the estimated response coefficients used to test such theories of the market-accounting relation are derived as elasticities from log-linear models, they will not provide a reliable indication of whether those theories should be confirmed or rejected.⁷

We first state the two main assumptions of the multiplicative theory of the long-run market-accounting relation. Direct formal implications are derived from the two assumptions. Direct empirical implications of the assumptions are tested in [Section 6](#).

Assumption A1: Market values and accounting values are ratio scales and the market-accounting relation is invariant to changes in those scales.

As noted earlier, all analysis using accounting numbers of the type considered in this paper implicitly assumes market and accounting values are measured on ratio scales (see footnote 2) and this assumption may therefore seem self-evident. However, it leads directly to a little-known result in the study of functional equations that has important implications for the form of the market-account relation. To assert that market and accounting values, M and A respectively, are both ratio scales means that they can both be transformed by the same positive scalar value, z , without changing the truth of statements using them. For instance, the statement that the market value of shares in a firm is twice that of its book value remains true whether the values are measured in millions of dollars, thousands of euros, or pounds sterling.⁸ Consequently, the valuation function F in $M = F(|A|)$ has the property that $F(z|A|) = Z(z)F(|A|)$, for any positive constant $Z(z)$ that is a function of z . Thus the market-accounting relationship does not depend on the currency or monetary unit used for measuring market and accounting values.

Below we demonstrate that $M = F(|A|) = \kappa|A|^\beta$ for some constants κ and β when market value is related to a single accounting variable. A proof of the multiplicative form of the relationship when more than one accounting variable is involved, of which the single variable

model is a special case, is a little more technical and is given in the Appendix. The constant z can be interpreted in this context as the rate of exchange between one currency and another.

By **A1**, $F(z|A|) = Z(z)F(|A|)$, where $z > 0$ and $Z(z) > 0$. Therefore, $F(z) = Z(z)F(1) = Z(z)\kappa$, denoting $F(1)$ as κ . Consequently, $Z(z) = \frac{F(z)}{\kappa}$ and $F(z|A|) = \frac{F(z)}{\kappa} F(|A|) = F(z) \frac{F(|A|)}{\kappa}$.

Defining $G(|A|) = \ln(\frac{F(|A|)}{\kappa})$, $\frac{F(|A|)}{\kappa} > 0$, so G is well defined:

$$\begin{aligned} G(z|A|) &= \ln\left(\frac{F(z|A|)}{\kappa}\right) = \ln\left(\frac{zF(|A|)}{\kappa}\right) = \ln\left(\frac{F(z)F(|A|)}{\kappa \cdot \kappa}\right) \\ &= \ln\left(\frac{F(z)}{\kappa}\right) + \ln\left(\frac{F(|A|)}{\kappa}\right) \\ &= G(z) + G(|A|) \end{aligned} \tag{5}$$

G obeys the same law as [Cauchy's \(1821\)](#) functional equality, $f(xy) = f(x) + f(y)$. The function $G(|A|)$ must therefore have the form $\beta \cdot \ln(|A|) = \ln(|A|)^\beta$ for some constant β . Using the definition of G above to translate it back to the valuation function F shows that $\frac{F(|A|)}{\kappa} = e^{G(|A|)} = e^{\ln(|A|)^\beta} = |A|^\beta$. Therefore, $F(|A|) = M = \kappa|A|^\beta$. In this context κ is to interpreted as the market to book ratio and β is the market elasticity with respect to $|A|$ in the non-linear relation $\kappa|A|^\beta$. From this it can be seen that the fundamental market accounting relation is invariant to z , the exchange rate, and takes the form of a power law.

The proof in the Appendix demonstrates that the power law principle carries over to products of the form $\prod_i |A_i|^{\beta_i}$. This result also extends to functions of accounting variables, A_i , $f(A_i)$, such that changes in the scales of the A_i can be factored out of f . Indexed linear-additive combinations of A_i , $(\sum_i b_i A_{i,t})^\beta$ have this property, since the market and accounting variables are measured on a common scale, so $Z(z) = z^\beta$ in that case. However, substituting $\sum_i b_i A_{i,t}$ into the single variable form of (1) illustrates the conceptual problems of interpretation, and beyond that, of estimation, caused by using additive linear models in trying to understand the market-accounting relation.

There are two possibilities. The first possibility is that the market elasticity on $|B_t| = |\sum_i b_i A_{i,t}|$ happens to be equal to 1, entailing that book value is sufficient to explain proportional changes in market value, i.e. market returns, absent any other information. When this occurs $M = |\sum_i b_i A_{i,t}|$. Consequently, when book value is positive and the $A_{i,t}$ are the fun-

damental variables in (2), the additive linear model without a constant term provides a valid mathematical form of the market-accounting relation to estimate. However this is only in the empirically unlikely event that $\beta = 1$, which is the same as assuming that book value does, in fact, completely explain market value. The second much more likely possibility is that the elasticity on book value is not equal to 1. Here, substituting $\sum_i b_i A_{i,t}$ for B_t in $M = \kappa |B_t|^\beta$ leads to $M = \kappa |\sum_i b_i A_{i,t}|^\beta$. Now, however, there is no algebraic way to identify the value of the individual response coefficients in the model and we cannot estimate the individual effects of the different account variables on market value.⁹

The multiplicative power law theory of the market-accounting relation expressed in (1) is tested empirically in Section 6. The hypotheses are expressed more formally as follows:

Hypothesis 1 (H1): The market-accounting relation has the form of the multiplicative power law shown in expression (1).

Hypothesis 1A (H1A): The market-accounting relation has the additive-linear form shown in (4).

The method of testing these hypotheses is explained in Section 4.2 below.

Assumption A2: The growth ratios of fundamental accounting variables, $g_i = \frac{|A_i|_t}{|A_i|_{t-1}}$ are stationary random variables, randomly sampled from the same joint distribution over time.

This is another way of stating [Gibrat's \(1821\)](#) law of proportionate effect. Most simply, since $|A_i|_t = g_i |A_i|_{t-1}$, if the growth variables are jointly distributed lognormally, $|A_i|$ is also jointly lognormal. More generally, if the growth variables are randomly sampled each interval of time a transaction occurs from the same, common, joint distribution, yearly observations of the $|A_i|$ approach the joint lognormal distribution as the number of transactions increases ([Kotz, Balakrishnan and Johnson, 2000](#); 20-28). Under these conditions, the $|A_i|$ are approximately jointly distributed according to a lognormal law in large samples of annual data.¹⁰

As the $|A_i|$ are approximately jointly lognormal, for all t , so too is $v \prod_i |A_i|^{\beta_i}$ for any constants v and β_i . Since $|B_{t-1}| = |A_0|$ is lognormal and $\prod_i |A_i|^{\beta_i}$ are geometric combinations

of the $|A_i|$, it implies that M is also lognormal. Additionally, since joint lognormality implies the marginal and conditional distributions of the $|A_i|$ are lognormal, the hypothesis and alternative hypothesis tested with respect to **A2** are as follows:

Hypothesis 2 (H2): Market values and the magnitudes of accounting variables are distributed as approximately lognormal in large samples of data.

Hypothesis 2A (H2A): Market values and accounting variables are distributed as approximately normal in large samples of data.

4.2 Method

We test hypothesis **H1** by comparing the OLS estimated coefficients from (1), i.e. from the single variable multiplicative model, $M_t = \kappa |B|_t^\beta$, and the multivariable multiplicative model, $M_t = v \prod_i |A_i|_t^{\beta_i}$, respectively, against the OLS estimated coefficients from their corresponding models, after dividing the model variables on both sides of the equality sign by $|B|_t$ for the single variable model and by $|B|_{t-1}$ for the multivariable model.

If the multiplicative power law form of the market-accounting relation in (1) is correct, simple algebra implies that in the single variable market to book model, $\frac{M_t}{|B|_t}$ equals $\kappa |B|_t^{\beta-1}$ and, in the multivariable market to book ratio model, $\frac{M_t}{|B|_{t-1}}$ (i.e. $\frac{M_t}{|A_0|}$) equals $v \prod_i |A_i|_t^{\beta'_i}$ where $\beta'_0 = \beta_0 - 1$ and $\beta'_i = \beta_i$ for $i \in \{1..4\}$.

Likewise, we test hypothesis **H1A** by comparing the OLS estimated coefficients in $M_t = k + bB_t$ and $M_t = v + \sum b_i A_i$ with the respective OLS estimated coefficients from the corresponding market to book ratio models, again dividing both sides by B_t in the single variable model and B_{t-1} in the multivariable model.

If the additive-linear form of market-accounting relation in (4) is correct, this implies that the single variable market to book ratio model should be $\frac{M_t}{B_t} = \frac{k}{B_t} + b$ and the multivariable market to lagged book ratio model should be $\frac{M_t}{B_{t-1}} = \frac{v}{B_{t-1}} + \sum_i b_i \frac{A_i}{B_{t-1}}$. In the case of the single variable model, hypothesis **H1A** can be tested by comparing the slope coefficient (b), estimated using the market value model, with the constant term estimated in the corresponding market to

book ratio model. Likewise, the constant term (k) in the market value model should be equal to the slope coefficient on $\frac{1}{B_t}$ in the market ratio model.

Similar considerations apply in the case of the additive-linear multivariable model. We can compare the slope coefficients b_i for $i \in \{1..4\}$ in both the market value and market to book ratio models. These should be equal. The constant term, v , in the market value model should equal the coefficient on $\frac{1}{B_{t-1}}$ in the corresponding market to book ratio model. The constant term in the multivariable market to book ratio model should equal the slope coefficient on B_{t-1} , b_0 , in the multivariable market value model. The results of these tests are reported in [Section 6](#).

Hypotheses **H2** and **H2A** are tested by comparisons of histograms and P-P plots of unlogged and logged data with the Gaussian distribution function. We use the Lilliefors test to characterise the level of approximation to the normal distribution ([Lilliefors, 1967](#)). The Lilliefors statistic is the maximum distance that the estimated cumulative distribution function of the data (F_n) reaches from the Gaussian distribution function: $Lilliefors = \max|F_n - \Phi|$. A value of 0.05 for Lilliefors test is interpreted as showing the maximum distance that the stated distribution is from the Gaussian is five percent. A Lilliefors statistic less than 0.05 signifies that the distribution is ‘approximately Gaussian’. $H2$ and $H2A$ are therefore rejected for the accounting variables, if their Lilliefors statistic relative to the Gaussian is greater than 0.05.

The use of absolute values in estimating the elasticities means that the sign on the elasticity shows the direction of change of the dependent variables relative to the independent variable. This is because market values are always positive and the elasticity of M with respect to any $|A|$ is $\frac{\partial M}{\partial |A|} \cdot \frac{|A|}{M}$. Consequently, when observations involving profit and losses are pooled to give estimates of elasticities on the magnitude of earnings, OLS estimation provides a weighted average of the market elasticities of profits and losses. This is no different from pooling observations of profits and losses in estimating (4) in data that is unlogged and the weighting principle is the same.

The weights for each parameter β_j , for data in the j^{th} partition, in the value of the parameter for the pooled data, β_p , are given by:

$$\beta_p = [\sum_j \mathbf{X}_j' \mathbf{X}_j]^{-1} [\sum_j \mathbf{X}_j' \mathbf{X}_j \beta_j] \quad (6)$$

The matrix $\mathbf{X}_j$ and parameter vector β_j relate to observations on the j^{th} accounting variable and its parameter estimate in the j^{th} partition of the pooled data. The weights are therefore based upon the sums of squares of the j^{th} partition of the data relative to the sums of squares in the pooled data.

5 Descriptive statistics of the data

[Table 2](#) provides summary statistics, with observations from Compustat over the period 1972 to 2025. The sample firms are U.S. firms with non-zero net income and non-zero book value.¹¹ Market values are from the end of the first quarter after year-end. The resulting sample contains 264,963 firm-year observations.

[\[Table 2 about here\]](#)

The second column of [Table 2](#) shows how these observations are distributed among the positive, zero and negative values of each accounting variable. There are 20,783 negative book value observations constituting 7.8 percent of the total. There are 91,674 loss observations being 34.6 percent of the total. There are more positive other income observations (180,815, 68.9 percent of the non-zero observations) than negative observations (81,459; 31.1 percent). Dividends were paid in just under one-half of the cases (107,825; 40.7 percent) and not paid in the remaining 157,138. The arithmetic mean ratios are shown in the third column of the bottom rows of [Table 2](#). For instance, the market to book ratio is 35.71. The mean values of the logged ratios in column eight, e.g. m , $r_{0.5}$ are exponentiated for comparison with the unlogged ratio. The standard errors of these ratios are close to zero. When the latter are calculated for each year there is little variation over the years, showing that they behave close to being constant. There is a reduction in skewness and kurtosis in the logged data, indicating an attenuation to the Gaussian distribution.

6 Test of Hypotheses

6.1 Hypotheses 1 and 1A

Hypothesis **H1** in the case of the single variable model is that the estimated coefficients in $M_t = \kappa |B|_t^\beta$ and $\frac{M_t}{|B|_t} = \kappa' |B|_t^{\beta'}$ are such that $\kappa = \kappa'$ and $\beta' = \beta - 1$, where κ is the market to book ratio ($\frac{M}{|B|}$). Table 3, columns 2 and 3, shows these estimates. Panel 1 in Figure 1 shows their evolution. The estimates are identical and their time sequences cannot be visually distinguished.

[Table 3 and Figure 1 about here]

Panel 2 of Figure 1 shows estimates of β and β' . The correlation between these two estimates is 100 percent. β' is exactly equal to $\beta - 1$. Hypothesis **H1**, applied to the single variable model, is therefore not rejected. We conclude that the market-book relation is log-linear.

In the case of the multivariable model, hypothesis **H1** is that the estimated coefficients in $M_t = v \prod_i |A_i|_t^{\beta_i}$ and $\frac{M_t}{|B|_{t-1}} = v' \prod_i |A_i|_t^{\beta'_i}$ are such that $v = v'$, $\beta'_0 = \beta_0 - 1$ and $\beta_i = \beta'_i$ for $i \in \{1..4\}$. We omit annual estimates to save space. Instead, we show their evolution in Figure 2. The estimates of v and v' are identical and their time sequences can hardly be distinguished visually (Panel 1). Panel 2 shows that β'_0 is very close throughout the period to $\beta_0 - 1$. The estimates of β_i and β'_i for $i \in \{1..4\}$ are also almost identical (Panels 3-6). The correlations between the sequence plots of these pairs of parameter estimates is 100 percent.

[Figure 2 about here]

We therefore do not reject hypothesis **H1** as it applies to the multivariable model. The multi-variable market-accounting relation is also log-linear and the market accounting relation takes the form of a multiplicative power law.

Columns 6 to 9 of Table 3 and Figures 3 and 4 show the results of testing the alternative hypothesis (**H1A**).

[Figures 3 and 4 about here]

The panels in Figure 3 parallel those in Figure 1. Panel 1 compares k in $M_t = k + bB_t$ with the coefficient on $\frac{1}{B_t}$ in $\frac{M_t}{B_t} = k' + \frac{b'}{B_t}$. These should be equal (i.e. $k = b'$) if the single variable market-book relation is additive-linear. Also the slope coefficient in the single variable market

value model (b) should be equal to the constant (k') in the corresponding market to book ratio model. Panel 2 of Figure 3 shows the sequence plots of estimates of these coefficients. The estimates in each of Panels 1 and 2 generally disagree. The sequence plots in each case are very different. The correlation coefficients between the relevant sequences are less than half those corresponding to the log-linear model. We therefore reject **H1A** in the case of the single variable model.

Figure 4 compares sequence plots of parameters in the multivariable, market value model, $M_t = v + \sum b_i A_i$ with the coefficients in the corresponding market to book ratio model $\frac{M_t}{B_{t-1}} = \frac{v'}{B_{t-1}} + \sum_i b'_i \frac{A_i}{B_{t-1}}$. Panel 1 compares v in the market value model with the coefficient b'_0 in the multivariable market to book ratio model. Panel 2 compares sequence plots of the response coefficient on B_{t-1} (b_0) in the multivariable, market value model with the constant term (v') in the multivariable, market to book ratio model. The remaining panels compare the remaining corresponding parameter estimates in the market value and market to book ratio models. If the additive-linear model is correctly specified, the sequence plots in each of the panels should be close to being identical, as they are in the panels in Figure 2. Each Panel in Figure 4 discloses the Pearson correlation coefficient between the sequences shown. Again, the pairs of sequence plots behave very differently over time. We therefore reject hypothesis **H1A** for the multivariable model.

The market-accounting relation is not well-described by an additive-linear model.

6.2 Hypothesis 2 and 2A

The second hypothesis (**H2**) is that market value and the fundamental accounting variables are approximately distributed as jointly lognormal. The alternative hypothesis (**H2A**) is that these variables have an approximately normal distribution.

PP plots comparing unlogged and logged market values are shown in Panels 1 and 2 of Figure 5. These serve as a benchmark for interpreting the distributional form of the accounting variables since it is known that market value approximates the lognormal rather than the Gaussian. The second (third) pair of panels show the corresponding PP plots for book value (net in-

come). The odd numbered panels show unlogged values of the variables. The even-numbered panels show the plots for logs of the absolute values of these variables.

[Figure 5 about here]

There is a clear and strong normalising effect from taking logs of all the variables shown. The attenuation to the Gaussian by taking logarithms is similar in all the variables used to estimate (1). This applies to estimates of conditional distributions from regressions of one accounting variable on combinations of the others.

Figure 6 displays histograms showing the strongly Gaussian form of the logs of the absolute values of variables, including those in Panels 2, 4 and 6 of Figure 5. The panels are based on pooled data. Standardized data based on annual cross-sections yields similar patterns. Histograms of the other variables used to estimate (1) are not shown but are all like those in Figure 6.

[Figure 6 about here]

Figure 7, Panel 1, shows a histogram of predicted market values from a log-linear regression of market value on the five accounting variables $(B_{t-1}, E_t, D_t, C_t, O_t)$. Panels 2-6 show the predicted values of each of these variables on the remaining four, derived from log-linear models. These five panels provide an indication of the conditional distributional form of the accounting variables. A joint lognormal distribution implies all its marginal and conditional distributions are lognormal.

[Figure 7 about here]

The evidence displayed in Figures 5 to 7 is that market value and the values of the accounting variables used in model (1) are closer to a lognormal than a Gaussian form.

Table 4 shows the results of gauging the extent of the approximation to the lognormal as described in Subsection 4.2. It summarizes comparisons of the Lilliefors measure quantifying the normalising effect of applying the logarithmic transformation.

[Table 4 about here]

On the left-hand side of this table, the unlogged variables have distance measures between 0.40 and 0.49 and are therefore rejected as being approximately Gaussian. The right-hand side of the table shows that neither logged market value nor any of the logged accounting variables

exceed the value of 0.05 in distance from the Gaussian distribution as measured by the Lilliefors statistic.

Based on this evidence we do not reject the hypothesis, **H2**, that fundamental accounting variables are approximately lognormal. However, the same evidence and the same criteria for testing leads to the rejection of **H2A**.

The distributional properties of the logs of the absolute values of the accounting variables demonstrated here justify OLS as an estimation procedure for log-transformed data. We therefore estimate the multiplicative function (1) with data transformed to logs of absolute values to produce a log-linear model suitable for use with OLS:

$$\begin{aligned}\ln(M) &= \ln(\kappa) + \beta \ln|B| + \varepsilon \\ \ln(M) &= \ln(v) + \sum_i \beta_i \ln|A_i| + \varepsilon\end{aligned}\tag{7}$$

where ε is a normally distributed error term.

7 Log-linear versus additive-linear model estimates

Table 5 shows the estimated parameters and adjusted R^2 s for (7) in cross-sections for each year and pooled. Those for the single variable model in closing book value (i.e. $M = \kappa|B|^\beta$) are shown in columns 2 to 4. The corresponding estimates for the multivariable model (i.e. $M = v \prod_i |A_i|^{\beta_i}$) are shown in columns 5 to 11, where column 11 sums the coefficients on logged absolute values of $B_{t-1}, E_t, D_t, C_t, O_t$.

[Table 5 about here]

The rows of the upper section of Table 5 show estimates for each year of the sample period. The bottom section of the table shows estimates based on pooled data and average values for the cross sections. The significance of nearly all the parameters is high, with some exceptions where a parameter estimate is close to zero. The t -statistics are therefore not shown; some of them are in the hundreds.

Column 11 of Table 5 shows that the sum of the β coefficients is usually close to 1. One condition for this is that the standard error of the accounting ratios is close to zero in large

samples and they act as though they are constants. In this case $\beta = \sum \beta_i$, so that if $\beta = 1$, $\sum \beta_i = 1$. This property is used to decompose the market to book ratio in [Section 8](#).

[Figure 8](#) summarizes the evolution in the parameter estimates from the log-linear model.

[Figure 8 about here]

With [Table 5](#), this figure shows the following. First, there is strong inverse relationship between the elasticity of book values and earnings. During economic downturns, the value relevance of earnings drops and the value relevance of book values increases ([Burgstahler and Dichev, 1997](#)). Second, the elasticities on book value, earnings, and common dividends dominate the other elasticities showing they are the key fundamental variables, as long believed in financial analysis. Third, on average over the sample period, 90 percent of long-run market returns are explained by changes in book value. The R^2 shows that book value explains on average 78 percent of the variation in market value. Fourth, the sum of the elasticities in the multivariable model in columns 6 to 11 of [Table 5](#) remain close to one in all years. Fifth, common dividends always have a positive elasticity whereas preference dividends are often negative. When zero dividend observations are excluded, the elasticity on dividends rises by around a third to a half. This shows common dividends are value relevant, possibly having a signalling role that positively affects market returns (compare to [Miller and Modigliani \(1961\)](#)).

The elasticities on book value and earnings in the models estimated in the paper are long-run associations of accounting with market values ([Hendry, 1995](#)). The time series patterns in these two elasticities therefore appear as state-contingent rotations between long-run book-value and earnings elasticities. In expansions, especially when inflation and interest rates are relatively contained, the long-run earnings elasticities are closer to the book-value elasticities, even surpassing the latter in 1999. The estimates suggest that downturns, recessions and high-interest-rate regimes are associated with a rotation toward book value and away from current earnings. This pattern is consistent with adverse macroeconomic conditions reducing the perceived persistence of current-period earnings and increasing the relative informational importance of balance-sheet measures associated with resilience, liquidation value and medium-run recovery.

This is evident in the major North American economic cycles. Around the 1973–75 recession, with recession dating confirmed by the NBER/FRED series ([Federal Reserve Bank of St. Louis \(n.d.\)](#)), the long-run coefficients move from 1973 ($\beta_{B_{t-1}}, \beta_E$) $\approx$ (0.46, 0.48) to 1975 (0.60, 0.33). The same rotation appears during the Volcker disinflation and recessions. The late 1970s and early 1980s were marked by high inflation and the sharp monetary tightening that began in October 1979 ([Federal Reserve History, 2013](#)). The Federal Reserve’s own history notes the shift to stronger anti-inflation policies, while the federal funds series records the exceptionally high rates reached during that period ([Board of Governors of the Federal Reserve System, n.d.](#)). In the estimates, 1980 still shows a high earnings elasticity (0.43) and lower book elasticity (0.40), but by 1982 the position had reversed ($\beta_{B_{t-1}}, \beta_E$) $\approx$ (0.50, 0.30), illustrating a strong macro-consistency in the change from earnings or flow dominance toward book-value or stock dominance.

The 2001 downturn displays a similar rotation. The NBER chronology dates a U.S. recession to 2001 ([Federal Reserve Bank of St. Louis, n.d.](#)), while contemporaneous FOMC statements document rapid monetary easing in response to weakening activity ([Board of Governors of the Federal Reserve System, 2001a,b](#)). The long-run elasticities move from ($\beta_{B_{t-1}}, \beta_E$) $\approx$ (0.26, 0.52) in 1999 to ($\beta_{B_{t-1}}, \beta_E$) $\approx$ (0.52, 0.27) in 2000, (0.65, 0.12) in 2001, (0.61, 0.14) in 2002, reverting to more commonly seen relative values, (0.49, 0.32), in 2003. That repeats an evident recession-profile pattern in the data of increasing book-value elasticities and decreasing earnings elasticities.

In the Global Financial Crisis, the recession dates are based on the NBER/FRED chronology ([Federal Reserve Bank of St. Louis, n.d.](#)). Unemployment also rose sharply during that period in the BLS unemployment series distributed by FRED ([U.S. Bureau of Labor Statistics, n.d.](#)). The long-run book-value elasticities in 2008–09 were higher than average at 0.61 and 0.57, while the earnings elasticities dropped to below average at 0.14 and 0.24. By 2010, the more usually observed relativities had been reasserted with ($\beta_{B_{t-1}}, \beta_E$) $\approx$ (0.49, 0.35). This is consistent with an environment in which balance-sheet resilience, capital adequacy, and write-down capacity matter more than current income.

In 2020, a recession year, the book-value elasticity falls to 0.41 and earnings elasticity rises. This is explained by the market values in this study being measured at the end of the first quarter after year-end. Fiscal year 2020 market values are therefore measured near the end of March 2021, after the market had fully rebounded. The two-month NBER recession (February to April 2020) does not enter the 2020 measurement window. The large size of the estimated 2020 intercept scale parameters ($\kappa = 8.32$, $\nu = 15.02$) is consistent with a growth and intangible premium spike, not a flight to balance-sheet anchors. The timing issue with the way market values are measured, is specific to 2020 because the recession in that year was uniquely brief (two months) and the rebound uniquely fast, so the depressed period fell outside the March-2021 window. For the longer downturns the opposite holds. The clearest illustration is the GFC. Fiscal year 2008 market values, measured at end-March 2009, occur almost exactly on the market bottom, so there the window captures the downturn rather than missing it. This is why 2008 shows the book rotation at 0.61.

The inflation-tightening cycle after the pandemic shows a similar mechanism. In 2021, the macro environment saw the abrupt recession followed by recovery ([Federal Reserve Bank of St. Louis, n.d.](#)); annual CPI inflation was still low in 2020 at about 1.23%, then rose to about 4.70% in 2021, 8.00% in 2022, falling back to 4.12% in 2023, and 2.95% in 2024 ([World Bank, n.d.](#)). In the long-run coefficients, 2021 sits at close to average(0.49, 0.38) . Then, after the 2022 inflation shock and tightening cycle, the coefficients rotate again toward book value, the pattern in 2022–24 being (0.60, 0.29), (0.62, 0.31), and (0.62, 0.28). Again, this is consistent with a higher-discount-rate environment in which investors discount near-term earnings more heavily and place more weight on the balance-sheet anchor.

The common-dividend elasticity appears to act as a secondary but informative macro signal. Although it is not the main measured association with market returns, the estimates are consistent, at least up to 2010, with the elasticity becoming more prominent during some episodes of elevated uncertainty. Before 2010, it is relatively high in 1987, 1991, 2008. This might be interpreted as a ‘quality-commitment’ channel. When macro conditions are noisy, payout capacity carries more information than in less volatile periods. After 2010, the association between dividends and market returns, and its possible role in this regard, appears to be dimin-

ished. Nevertheless, the elasticities are always positive, as noted above, and this supports some kind of signaling role for common dividends. In contrast, the long-run preferred-dividend elasticity is mostly negative, which is consistent with it playing a more financing-structure-like role rather than it being a broad macro signal. This, as with all the observations concerning the patterns in the other elasticities above, is one possible interpretation of what these measures show regarding associations between investor reactions in financial markets and macroeconomic conditions, and requires separate empirical investigation.

The result of using the additive-linear model to estimate cross-section market response coefficients from 1972-2025 is shown in [Figure 9](#).

[Figure 9 about here]

The response coefficients in Panel 1 of this figure are highly volatile compared to the elasticities shown in [Figure 8](#), making them difficult to interpret both in the context of background economic events and with respect to each other. Removing the coefficient on preference dividends, which is the most volatile of the coefficients, in Panel 2, still leaves the appearance of high volatility and does not greatly help improve the understandability of the estimates. When shown separately, as in Panel 3, the coefficient on book value in this model is also volatile.

8 Decomposition of the market to book ratio

Compared to the additive-linear model, the multiplicative model enables the market to book ratio to be related more accurately to the main accounting return ratios with respect to book value and the response coefficients on the relevant accounting variables. Writing the right-hand side of (1) in terms of accounting ratios and letting r_i represent $\frac{|A_i|_t}{|B|_t}$,

$$\kappa |B|_t^\beta = v \left(\prod_i |r_i|^{\beta_i} \right) |B|_t^{\sum \beta_i} \quad (8)$$

Dividing M_t in (1) by $|B|_t$ produces the average market to book ratio, $\mu = \frac{M_t}{|B|_t}$. Using multiplicative theory, the estimate of μ based on the second term in (1), μ' , is $\mu' = \kappa |B|_t^{\beta-1}$.

Consequently, κ equals $\mu' |B|_t^{1-\beta}$. Substituting this equality for κ in (8) and rearranging gives a second estimate of μ , μ'' , based on the estimated multivariable model in (1),

$$\mu'' = v \left(\prod_i |r_i|^{\beta_i} \right) |B|_t^{\sum \beta_i - 1} \quad (9)$$

This gives, for all t , an estimate of μ , as a function of book value, the elasticities on the accounting variables (β_i) and the ratios of the accounting variables with respect to book values (r_i).

Estimates of μ , μ' and μ'' , based upon substituting the geometric averages for the variables in $\mu = \kappa |B|_t^{\beta-1}$ (μ') and in (9) (μ''), are shown in columns 2–4 in Table 6, respectively. Their five percent confidence intervals overlap and the untabulated correlations between μ and μ' and μ'' respectively are 100 percent and 99.1 percent. The time sequences of the estimates are shown in Panel 1 of Figure 10 and display this close correspondence.

[Table 6 and Figure 10 about here]

Consequently, the relationship shown in (9) between the market to book ratio, the main accounting ratios and the accounting variable response coefficients, i.e. elasticities, is seen to be a reasonably precise way of tracing movements in the market to book ratio to its fundamental accounting causes.

When $m = \frac{M}{B}$, the market to book ratio based on the additive-linear model, $M = a + bB$, is $m' = \frac{k}{B} + b$ and the formula corresponding to (9) is

$$m'' = \frac{v}{B_t} + \sum_i b_i r_i. \quad (10)$$

A comparison of the estimates of m , m' , and m'' , this time by substituting arithmetic averages for the accounting variables and ratios, is shown in columns 5-7 in Table 6 and in Panel 2 of Figure 10. The untabulated correlations between the value of the market to book ratio and its estimates from the single variable and multivariable additive-linear models are 13.1 percent and –67.8 percent respectively and their five percent confidence intervals do not overlap.

9 The effects of pooling logs of absolute values

Table 7 shows the effect of pooling data on estimates of the elasticities. We partition the data into subsets based on the signs of the untransformed accounting variables. We restrict the set to observations with non-zero earnings values and focus on positive book values, which constitute the large majority in the data set. Dividends can only be zero or positive. The other variables can be positive, zero, and negative. This restricts the number of partitions considered in Table 7 to $1 \times 2 \times 2 \times 2 \times 3 = 24$ for accounting variables B_{t-1} , E_t , D_t , C_t , and O_t .

[Table 7 about here]

The rows in Table 7 show estimated parameters relating to the variables in the multivariable model in (1) where the variables have the indicated signs and zeros in columns 2-7. For example, in row 11, the data used for estimation of (1) consists of cases with positive book values, positive earnings, and zeros for the other variables. Below row 24, the table pools observations with negative book values. Some partitions are not reported because of a lack of observations, preventing the coefficients to be estimated, i.e. partition 14. The missing partition is where the O variable is zero.

The partitions reveal how elasticities are affected by the signs on the fundamental accounting variables. The elasticities generally sum close to 1, see column $\sum_i \beta_i$. There are some exceptions, particularly in those instances where earnings are negative, e.g. for partition 23.¹²

The elasticity of book value is generally higher than that of earnings and the elasticities of the latter have either negative signs or low positive signs in cases where earnings are negative. When dividends are paid, their market elasticity is positive and about 19 percent. This compares to the overall elasticity of 14 percent for common dividends shown in Figure 8. The pattern of elasticities in Table 7 is consistent with the time series patterns shown by the cross-section models and reinforces the fact that book values, earnings and dividends are the accounting variables having the strongest association with market returns.

10 Conclusion

We demonstrate that the market-accounting relation follows the form of a multiplicative power law between firm market value and the magnitude of fundamental accounting values. A special case of this result is that the relation between market and book value also takes the form of a power law. Thus the best form for estimating the market-accounting relations by ordinary least squares is log-linear. Furthermore, the magnitudes of accounting variables are approximately distributed as lognormal so that ordinary least squares estimates based on log-linear models can be tested with confidence using standard inferential methods.

Two assumptions underlie this theory of the relation between market values and accounting variables. The first assumption is that market and accounting variables are measured on a continuous ratio scale and that the market-accounting ratio is invariant to changes of scales in those variables. The second assumption is that accounting variables are generated by an exponential growth process. These are the only substantive assumptions needed to produce the results reported in this paper.

The power law form of the market-accounting relation shows that elasticities are the most useful accounting response coefficients for analyzing changes in market value associated with changes in accounting values. The power law produces response coefficients immune to changes of scale measuring the market and accounting variables. Measures of proportional change makes elasticities the best measure to describe representative response coefficients in samples of data, as market and accounting values often vary greatly in size from sample to sample. Since elasticities are constant in the correct power law specification of the market-accounting relation, changes in the sizes of market and accounting variables forces the instantaneous linear rate of response of market value to changes in the accounting variables, which is the limit of the coefficients estimated in the additive-linear model, to change with each sample of data. This adds another source of unwanted spurious variability to coefficient estimates: in addition to model misspecification, the parameters in the additive-linear model vary with any change in the constituents of samples. This explains the excessive volatility in coefficient estimates from the additive-linear model, both in this paper and in the literature generally.

Conceptually the multiplicative power law provides an interpretation of market value as a volume measure based on a cube in hyperspace. The length of each side of the cube is given by an accounting variable weighted by its elasticity and the entire cube is multiplied by a factor representing the market value of intangible components of the accounting variables. This is illustrated in [Figure 11](#) for three accounting variables: capital contributed to date, returns (e.g. earnings for the current accounting period) and distributions (e.g. dividends for the current accounting period).

[Figure 11 about here]

It is the multidimensional property of the multiplicative model that allows the separate response effects of these three accounting dimensions on market value to be validly and reliably estimated. No constraints are imposed on its parameter values. In contrast, the linear-additive model is uni-dimensional, representing firm value as the length of a straight line given by the linear combination of the accounting variables described in that model. The separate response effects of the accounting variables can only be extracted if the overall market elasticity of response of the linear combination is 1. This is equivalent to making the assumption, absent any empirical evidence, that the additive linear model fully explains market returns. When this forceful and very unreasonable assumption is not true, the response coefficients in the additive-linear model are neither valid nor reliable and do not reflect the separate effects of their different variables on returns.

Appendix: Proof of multiplicative power law form of the market-accounting relation.

Definition: Let $\odot$ be the point-wise product of two vectors $\mathbf{x}$ and $\mathbf{y}$, of equal length i.e.,

$(x_i) \odot (y_i) = (x_i y_i)$. If a vector is of length 1 then $\mathbf{x} = (x) = x$, $\mathbf{y} = (y) = y$ and $\mathbf{x} \odot \mathbf{y} = (x) \odot (y) = (xy) = xy$. In general, $\mathbf{x} \odot \mathbf{y} = (x_1, x_2, \dots, x_n) \odot (y_1, y_2, \dots, y_n) = (x_1 y_1, x_2 y_2, \dots, x_n y_n)$. If $\mathbf{c}$ is a vector of n elements all having the same scalar value c and $\mathbf{x}$ is a vector of variables of the same length, then $\mathbf{c} \odot \mathbf{x} = \mathbf{cx}$. ■

Lemma: If $\varphi(\mathbf{x})$ is a positive scalar valued function of positive vectors $\mathbf{x} \equiv (x_1, x_2, \dots, x_n)$ such that $\varphi(\mathbf{x} \odot \mathbf{y}) = \varphi(\mathbf{x}) + \varphi(\mathbf{y})$ then there exists a vector $\mathbf{k}$ of length n such that

$\varphi(\mathbf{x}) = \mathbf{k} \cdot \ln[\mathbf{x}]$, where $\ln[\mathbf{x}] \equiv (\ln[x_1], \ln[x_2], \dots, \ln[x_n])$.

Proof: Assume $\varphi(\mathbf{x} \odot \mathbf{y}) = \varphi(\mathbf{x}) + \varphi(\mathbf{y})$ and define $\psi(\mathbf{x}) \equiv \varphi(e^{\mathbf{x}})$ where the exponential operator operates, as with the logarithmic operator, point wise on the elements of the vector. Then $\psi(\mathbf{x} + \mathbf{y}) = \varphi(e^{\mathbf{x} + \mathbf{y}}) = \varphi(e^{\mathbf{x}} e^{\mathbf{y}}) = \varphi(e^{\mathbf{x}}) + \varphi(e^{\mathbf{y}}) = \psi(\mathbf{x}) + \psi(\mathbf{y})$.

Then there exists a vector of constants $\mathbf{k}$ of length n such that $\psi(\mathbf{x}) = \mathbf{k} \cdot \mathbf{x}$ (Eichhorn, 1978). For positive values of the function ψ and positive vectors $\mathbf{x}$ and $\mathbf{y}$, $\varphi(\mathbf{x}) = \psi(\ln[\mathbf{x}])$. Hence a vector $\mathbf{k}$ exists such that $\varphi(\mathbf{x}) = \mathbf{k} \cdot \ln[\mathbf{x}]$. ■

Theorem: Let the elements of vector $A = (a_1, a_2, \dots, a_n)$ be ratio scale measures of the absolute value of additive accounting variables. F is a positive scalar valued, vector function of market values of $\mathbf{A}$ such that $F(z\mathbf{A}) = zF(\mathbf{A})$ where z is a positive exchange rate with respect to a numeraire currency or, more restrictively, with respect to different units of measurement within a single currency (e.g. cents, dollars and millions of dollars). Then constants v and β_i , $i = 1$ to n exist such $F(\mathbf{A}) = v \prod_i A_i^{\beta_i}$, $v > 0$.

Proof: Let $\mathbf{1}$ be a vector of length n whose elements are all 1. The market value of a vector of unit measures of the accounting variables, $F(\mathbf{1})$, is such that $F(z\mathbf{1}) = zF(\mathbf{1}) = F(\mathbf{z})$. Thus $z = \frac{F(\mathbf{z})}{F(\mathbf{1})}$.

Define $\eta(\mathbf{A}) = \ln\left[\frac{F(\mathbf{A})}{F(\mathbf{1})}\right]$. Then,

$$\begin{aligned}
\eta(\mathbf{z} \odot \mathbf{A}) &= \ln[F(\mathbf{z} \odot \mathbf{A})/F(\mathbf{1})] \\
&= \ln[(zF(\mathbf{A})) / F(\mathbf{1})] && \text{(by definition.)} \\
&= \ln[(F(\mathbf{z})/F(\mathbf{1}))[F(\mathbf{A}) / F(\mathbf{1})]] && (11) \\
&= \ln[(F(\mathbf{z})/F(\mathbf{1})) + \ln[F(\mathbf{A}) / F(\mathbf{1})]] \\
&= \eta(\mathbf{z}) + \eta(\mathbf{A}).
\end{aligned}$$

By the Lemma above, therefore, a vector β of length n exists such that $\eta(\mathbf{A}) = \beta \cdot \ln[\mathbf{A}]$. Since $F(\mathbf{A}) / F(\mathbf{1}) = e^{\eta(\mathbf{A})}$, $F(\mathbf{A}) = F(\mathbf{1})e^{\eta(\mathbf{A})} = \nu e^{\eta(\mathbf{A})}$, where $F(\mathbf{1}) \equiv \nu$. Thus $F(\mathbf{A}) = \nu e^{\beta \cdot \ln[\mathbf{A}]}$ for some vector β of length n . Hence, $F(\mathbf{A}) = \nu e^{\sum_i \ln A_i \beta_i}$ for each element i in β and $\mathbf{A}$. Consequently, $F(\mathbf{A}) = \nu \prod_i A_i^{\beta_i}$, where $A_i, \nu > 0$. ■

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Notes

¹The estimated response coefficients reported in this paper are based upon models without lagged variables. The estimates are therefore of long-run, not short-run effects, see [Madsen \(2005\)](#).

²It would not be possible to make such statements if categorical, ordinal or interval scales were used to measure market and accounting variables. Further, the unit of measurement, cents, dollars or millions of dollars, has to be known to make sense of both market and accounting numbers, so their measurement cannot be based on a nominal scale either.

³Both κ and ν can be interpreted as measures of intangible value contained in accounting variables or, equivalently, as measures associated with the market's perception of riskiness in current share prices.

⁴Other accounting variables that add in value to closing book value could be used. We refer to any accounting variables satisfying this condition as 'fundamental' accounting variables.

⁵The ratio-scaled properties of cost-values and market-values are derived from their method of construction by accountants, see [Willett \(2026\)](#).

⁶An alternative exposition in a more general mathematical context is given in [Eichhorn \(1978\)](#).

⁷Estimates of response coefficients based on the additive-linear model do not clearly reveal the value relevance or otherwise of common dividends, for instance.

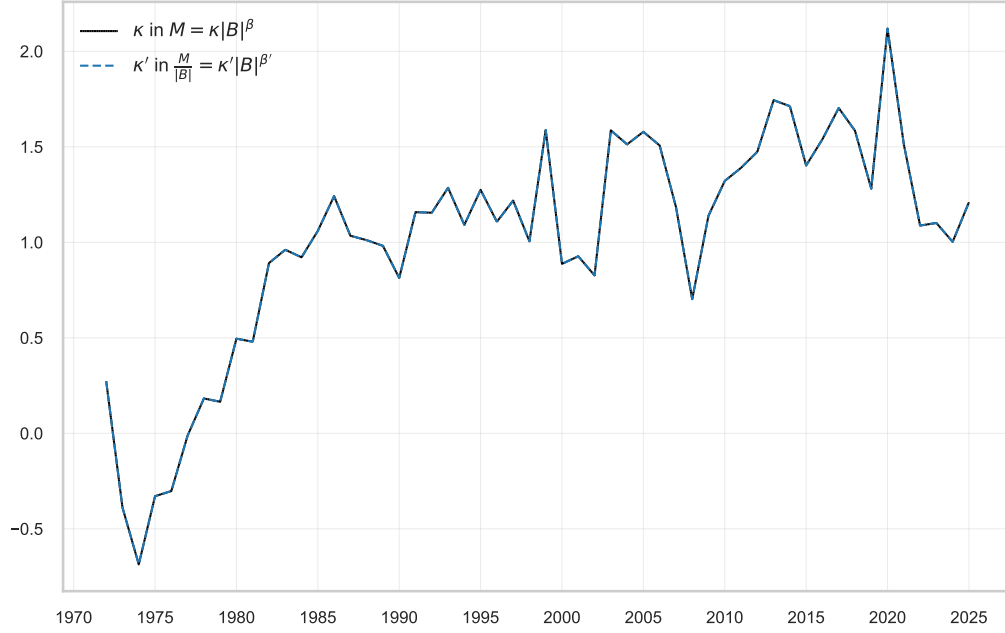
⁸This argument follows [Roberts \(1979\)](#) based on [Luce \(1959\)](#); [Aczél \(1966\)](#) and [Cauchy \(1821\)](#).

⁹E.g. there is no general algebraic solution, (c, d) , to the equation $(ax + by)^\beta = cx + dy$, when β is not an integer.

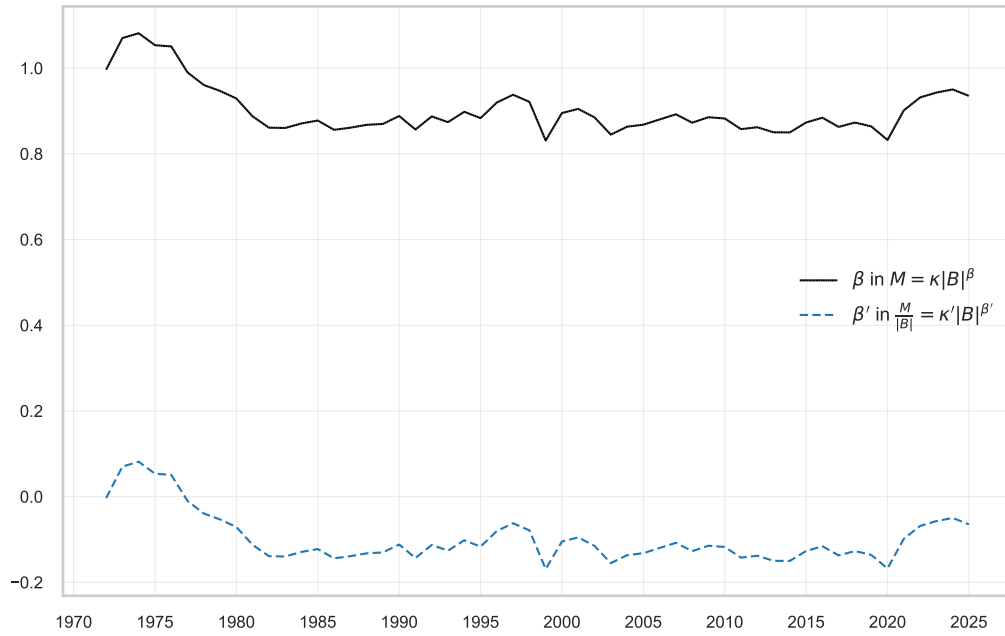
¹⁰In the rest of the paper we leave out the t subscripts unless the context requires us to do otherwise.

¹¹These restrictions currently have a negligible effect on the sample, excluding no additional observations.

¹²However, the number of observations for this partition is 1,146 over a sample period of 54-years.

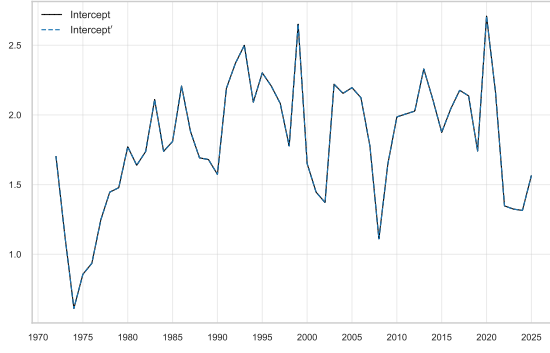


(Panel 1) Estimates of κ in $M = \kappa|B|^\beta$ (black) and κ' in $\frac{M}{|B|} = \kappa'|B|^{\beta'}$ (blue).

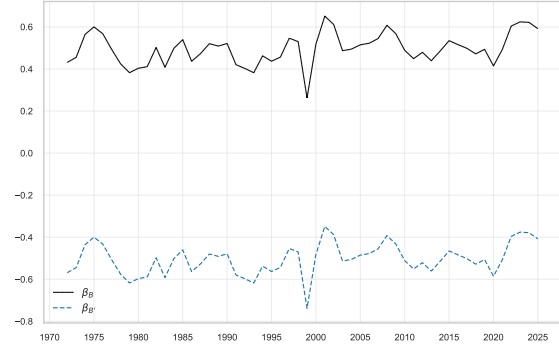


(Panel 2) Estimates of β in $M = \kappa|B|^\beta$ (black) and β' in $\frac{M}{|B|} = \kappa'|B|^{\beta'}$ (blue).

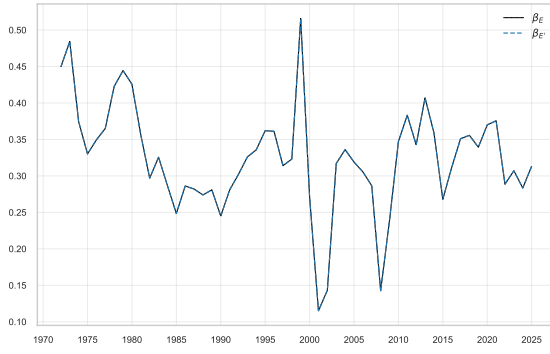
Fig 1: Test of hypothesis **H1**



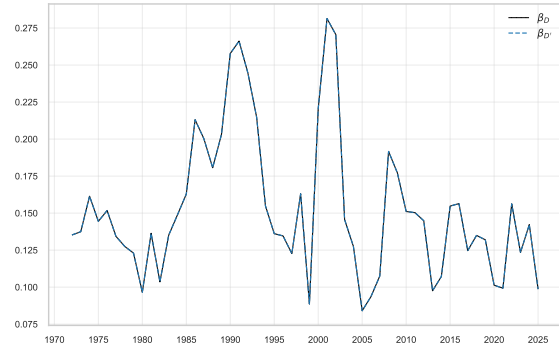
(Panel 1) Intercept (v)



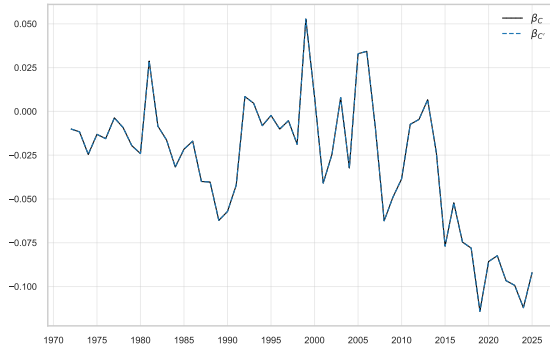
(Panel 2) Book Value (A_0 or B_{t-1})



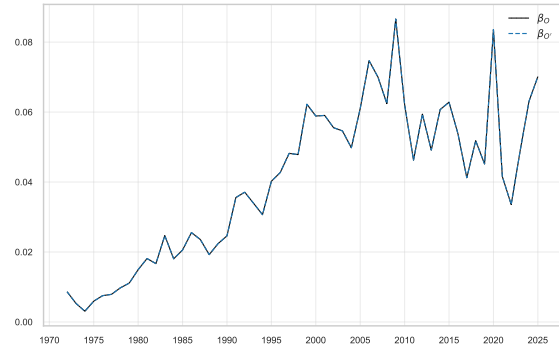
(Panel 3) Earnings (A_1)



(Panel 4) Dividend (A_2)

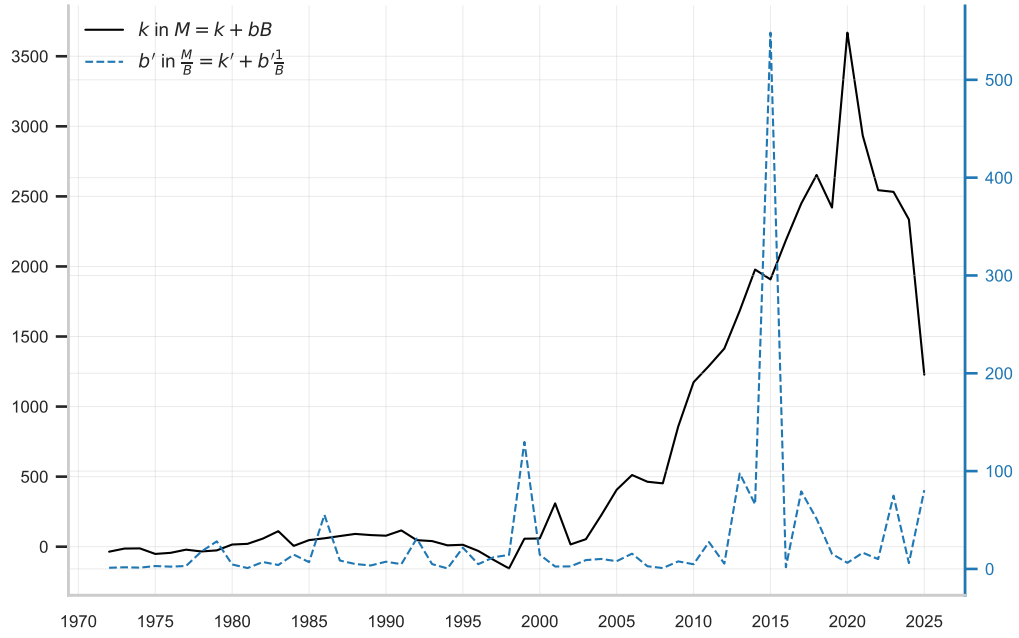


(Panel 5) Preferred dividend (A_3)

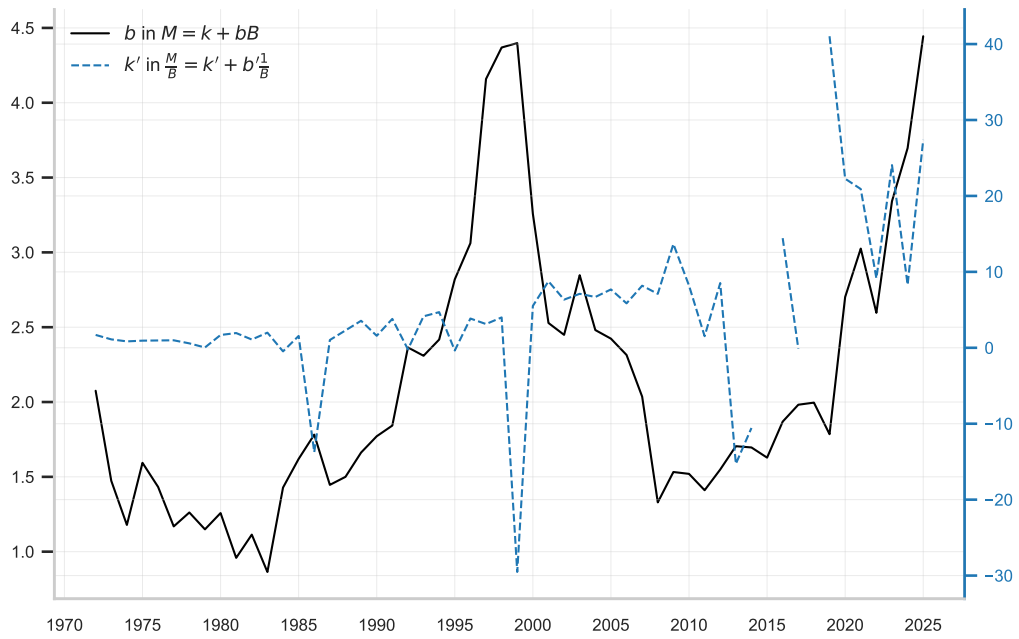


(Panel 6) Other (A_4)

Fig 2: Estimates of $M_t = v \prod_i |A_i|_t^{\beta_i}$ (black) and $\frac{M_t}{|B|_{t-1}} = v \prod_i |A_i|_t^{\beta'_i}$ (blue).

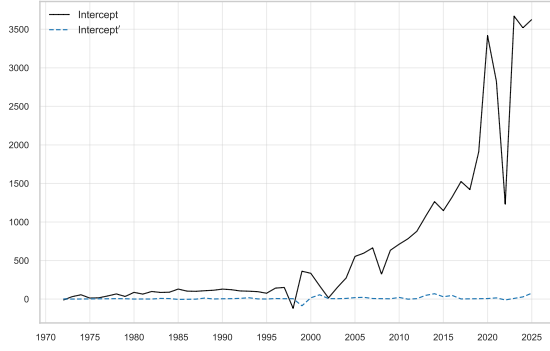


(Panel 1) Estimates of $M = k + bB$ and $\frac{M}{B} = k' + b'\frac{1}{B}$ (blue)

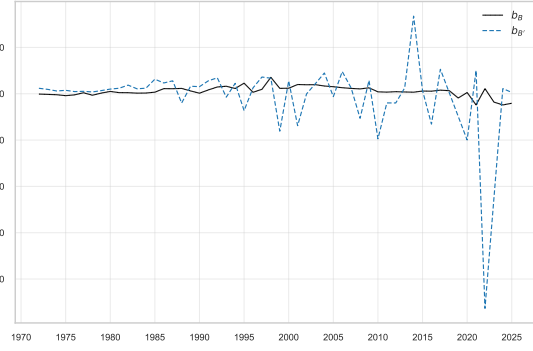


(Panel 2) Estimates of $M = k + bB$ and $\frac{M}{B} = k' + b'\frac{1}{B}$ (blue)

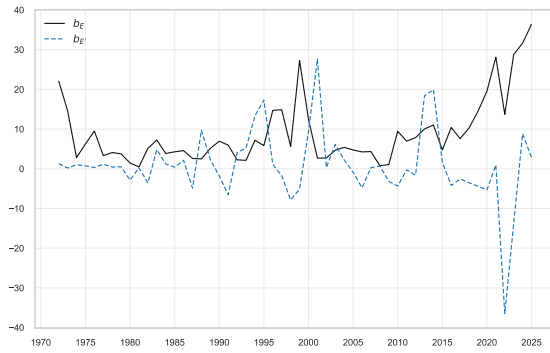
Fig 3: Test of hypothesis **H1A**



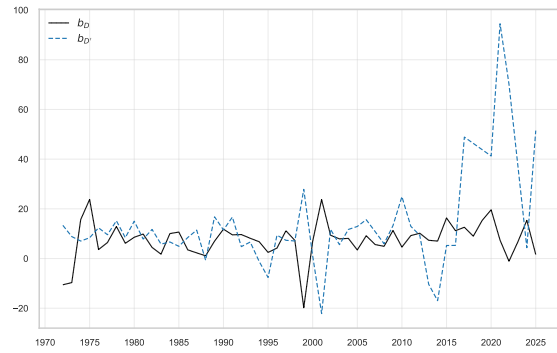
(Panel 1) Intercept ($v, \rho = 50.1\%$)



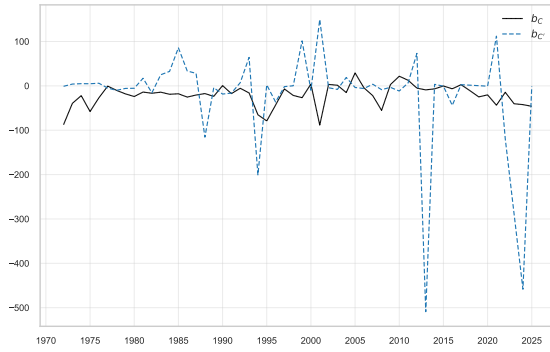
(Panel 2) Book Value ($A_0, \rho = 11.1\%$)



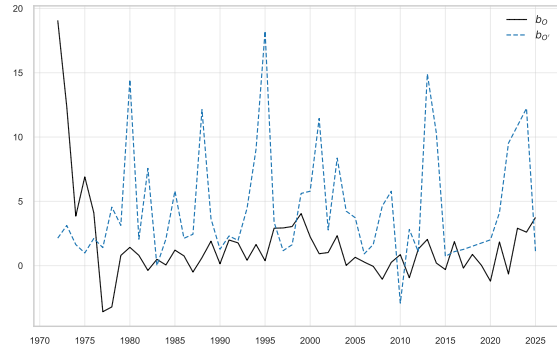
(Panel 3) Earnings ($A_1, \rho = 10.0\%$)



(Panel 4) Dividend ($A_2, \rho = -22.7\%$)



(Panel 5) Preferred dividend ($A_3, \rho = 30.8\%$)



(Panel 6) Other ($A_4, \rho = 33.9\%$)

Fig 4: Estimates of $M_t = v + \sum b_i A_i$ (black) and $\frac{M_t}{B_{t-1}} = \frac{v}{B_{t-1}} + \sum_i b_i \frac{A_i}{B_{t-1}}$ (blue).

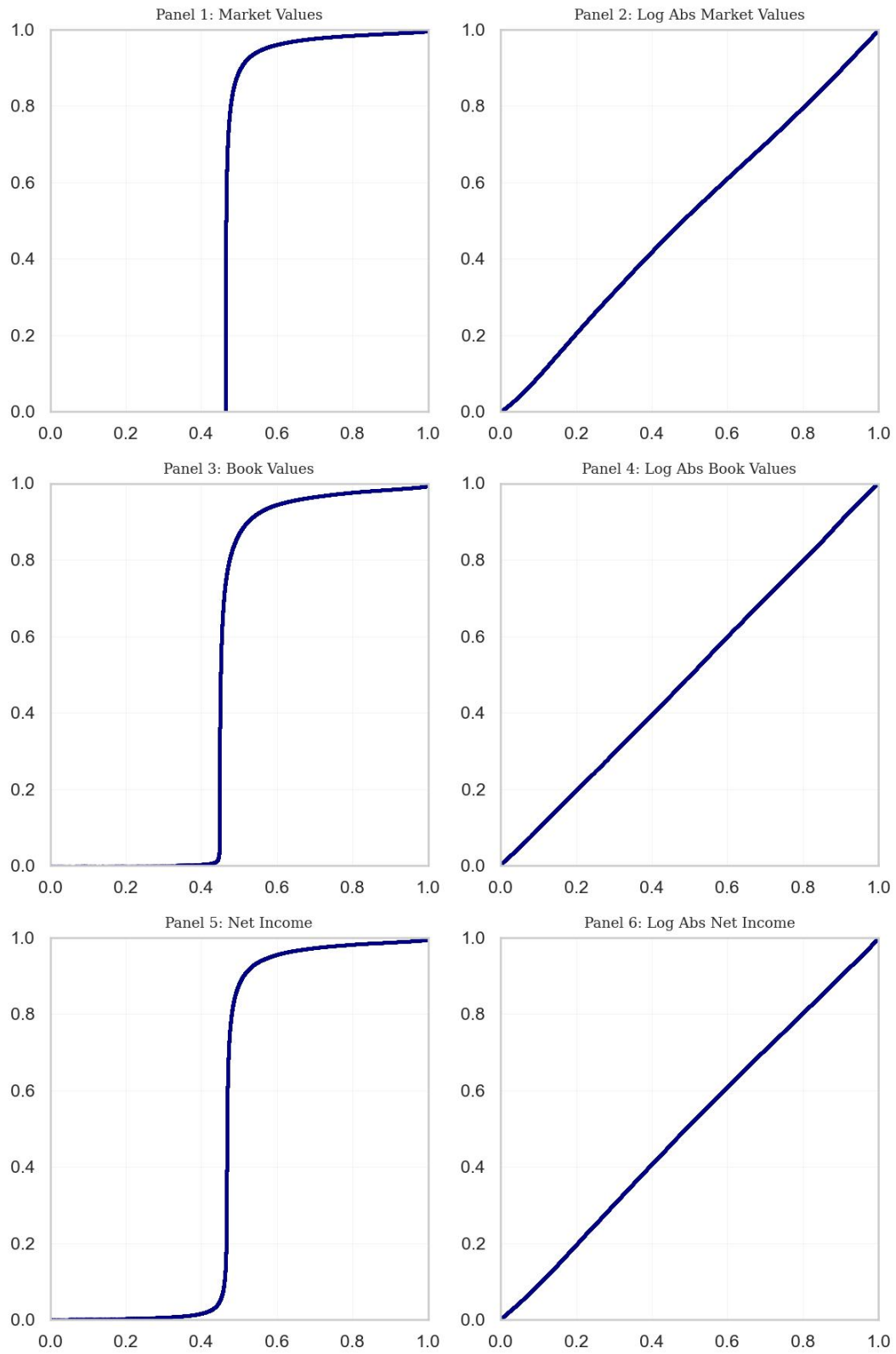


Fig 5: Compustat firms 1971-2016 – Normal PP Plots – Fundamental Variables

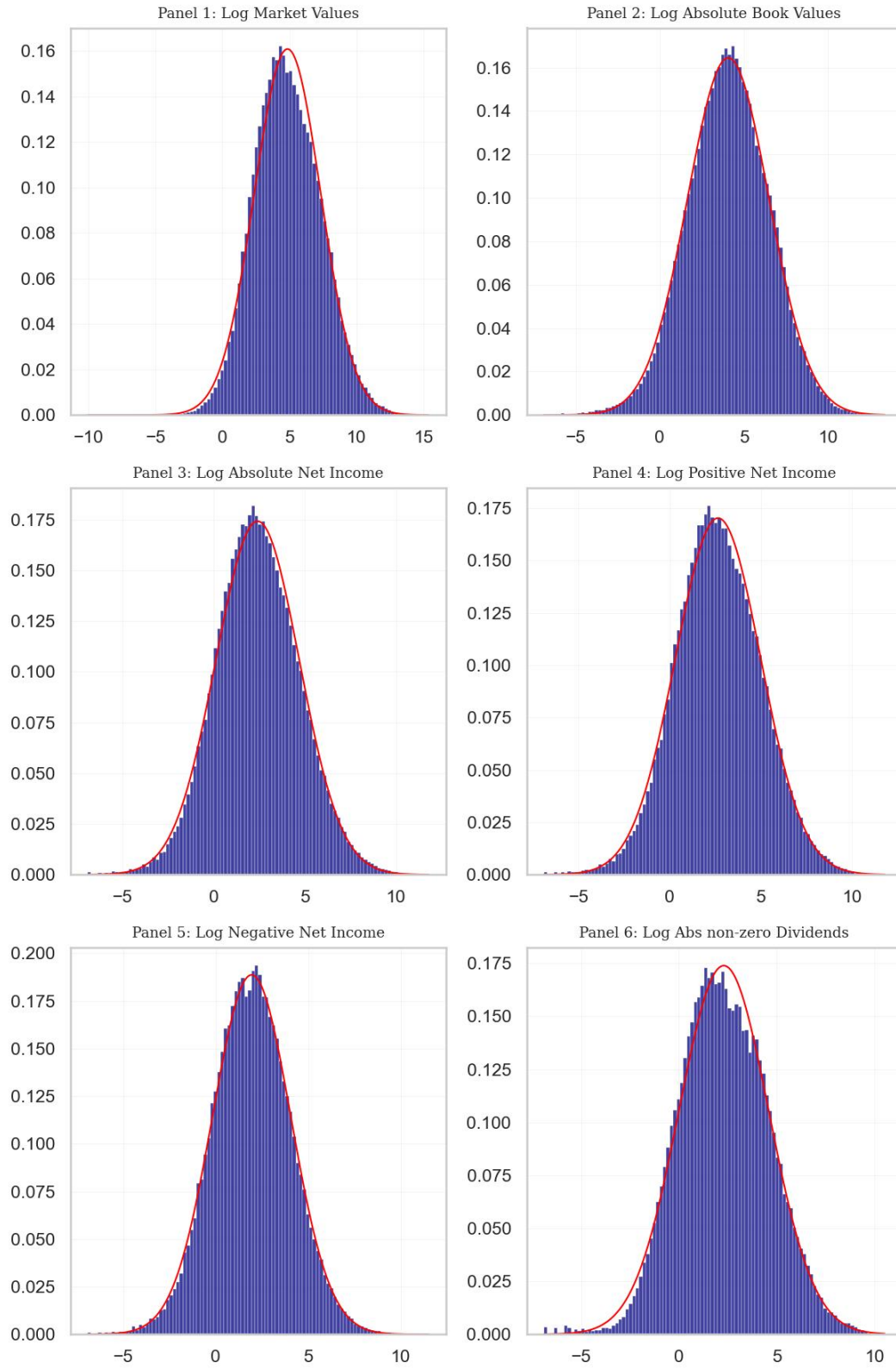


Fig 6: Histograms of Key Fundamental Accounting Variables in Logged Data

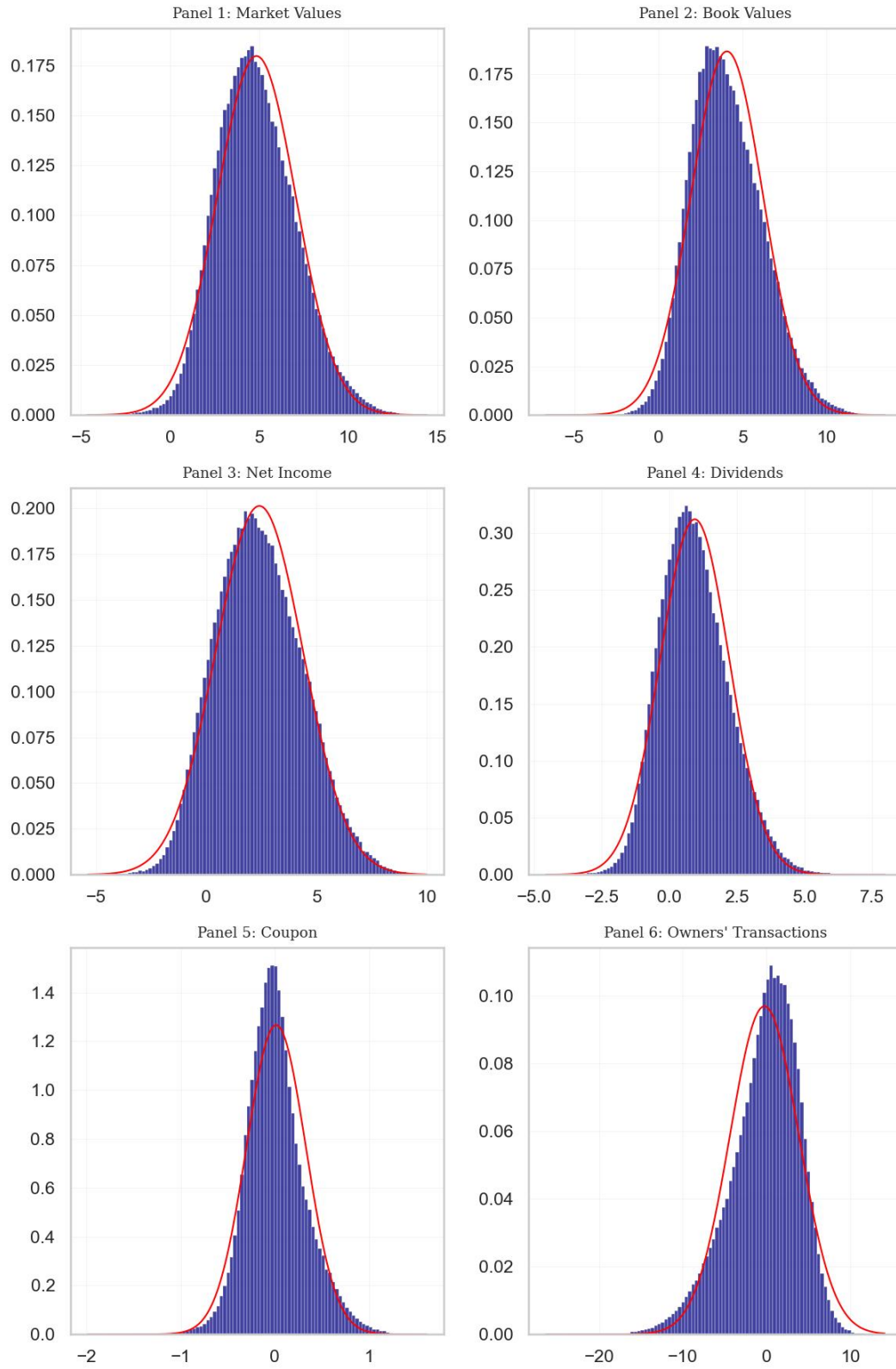


Fig 7: Histograms of Predicted Values based on accounting variables in logs.

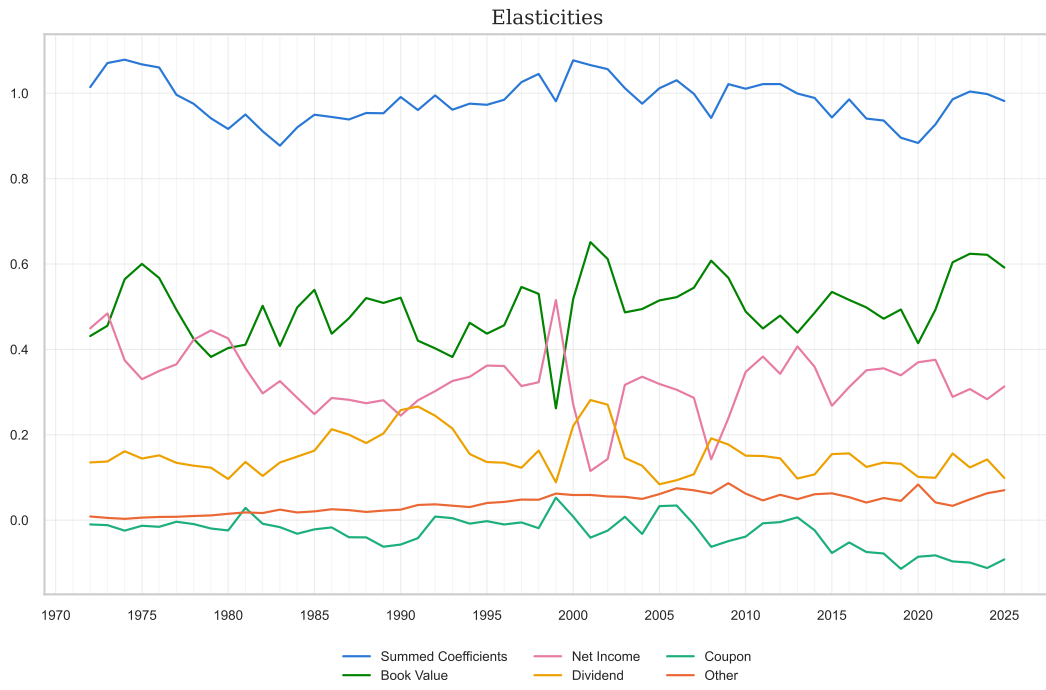
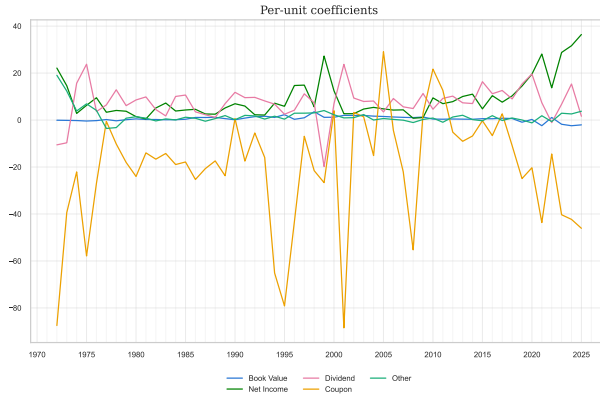
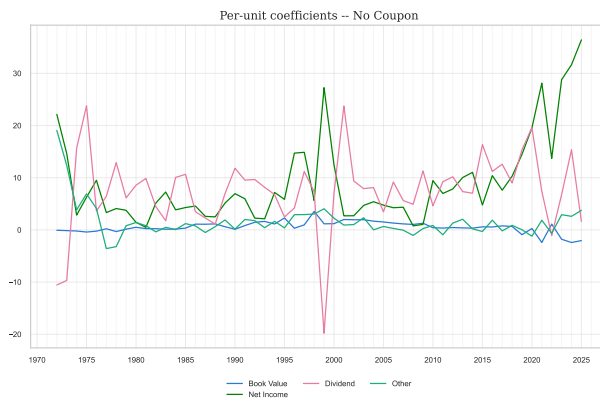


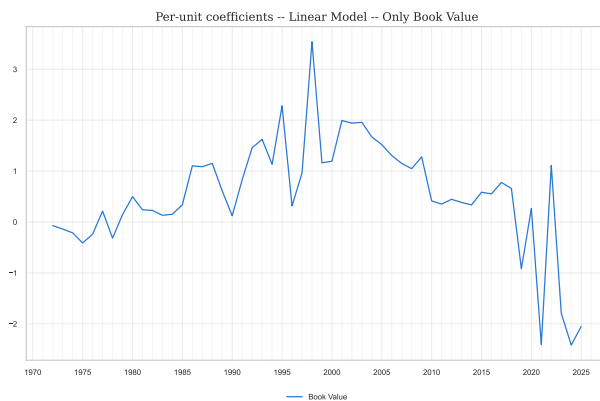
Fig 8: Empirically estimated elasticities using $M = v \prod_i |A_i|^{\beta_i}$. See Columns 6 to 11 of [Table 5](#).



(Panel 1) Per-unit coefficients estimated using the additive-linear model.

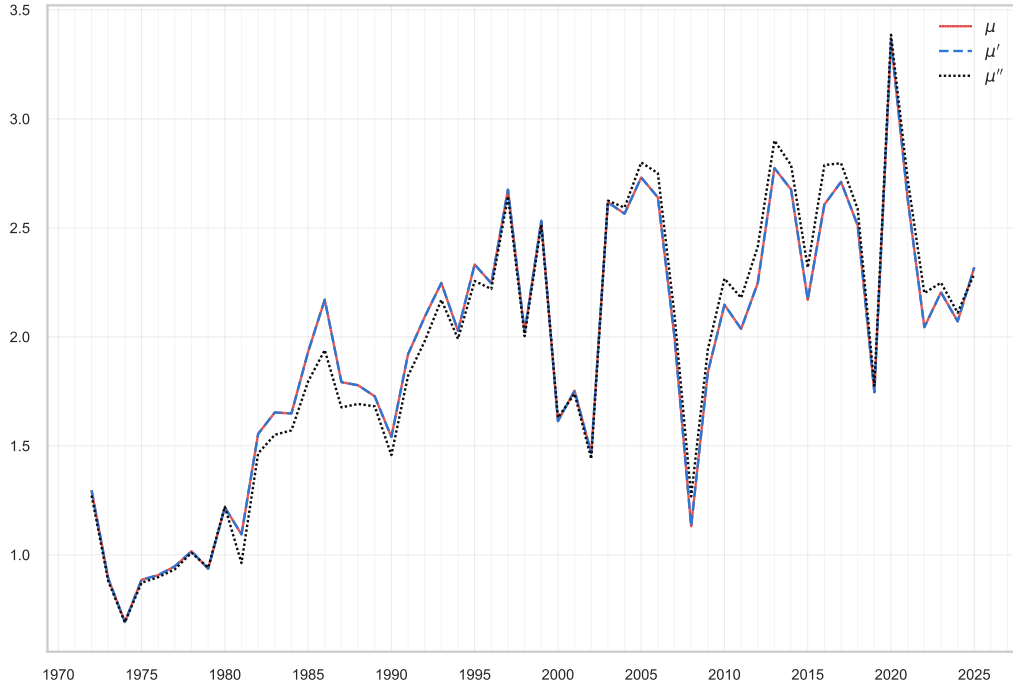


(Panel 2) Per-unit coefficients estimated using the additive-linear model, excluding preference shares.

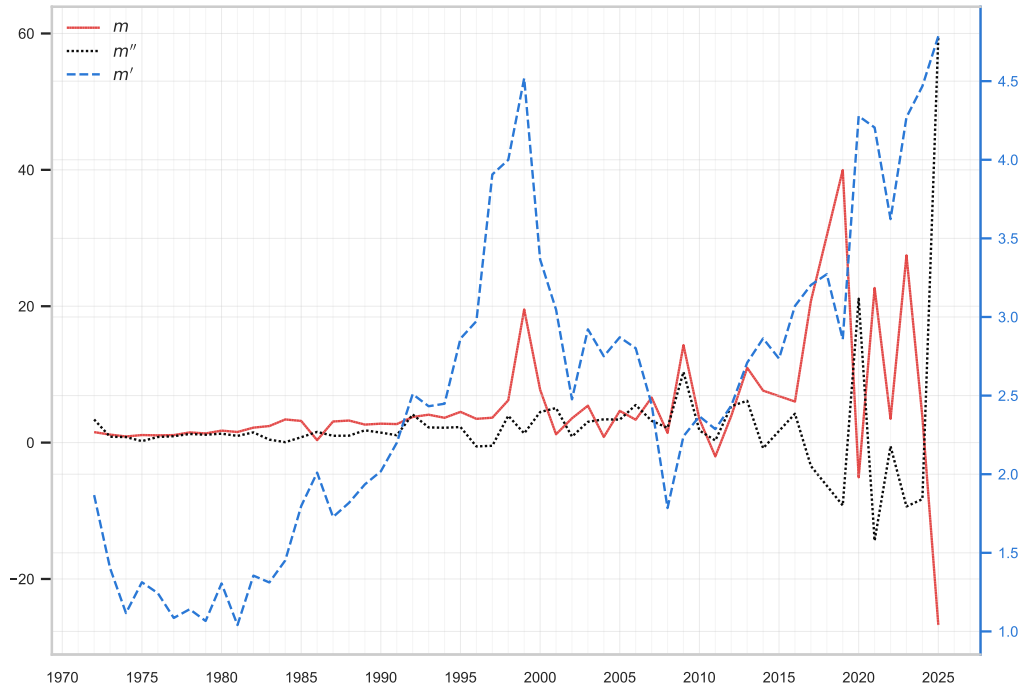


(Panel 3) Per-unit coefficients estimated using the additive-linear model, book value response coefficients.

Fig 9: Empirically estimated coefficients predicted by a additive-linear model, e.g. $M_t = v + \sum b_i A_i$.

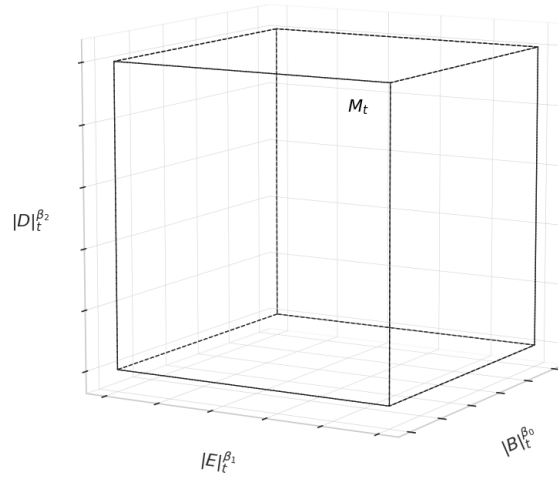


(Panel 1) Empirically estimated $\mu = \frac{M_t}{|B|_t}$, $\mu' = \kappa |B|_t^{\beta-1}$, and $\mu'' = v \left(\prod_i |r_i|^{\beta_i} \right) |B|_t^{\sum \beta_i - 1}$, see columns 2-4 of Table 6. The variables are geometric means.

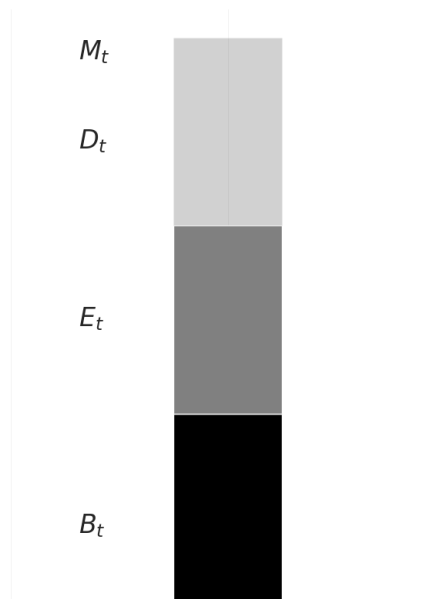


(Panel 2) Empirically estimated $m = \frac{M}{B}$, $m' = \frac{k}{B} + b$, and $m'' = \frac{v}{B_t} + \sum_i b_i r_i$, see columns 5-7 of Table 6. The variables are arithmetic means.

Fig 10: Decomposition of the market to book ratio.



(Panel 1) Market value as a volume measure based on a cube in hyperspace. M_t is market value of the firm, B_t is capital, E_t is returns, and D_t is distributions.



(Panel 2) M_t is market value of the firm, B_t is capital, E_t is returns, and D_t is distributions.

Fig 11: The multi-dimensional property of the multiplicative model.

TABLE 1
Variables used in models

Symbols used in expressions	Compustat Xpressfeed (Legacy code)	Definition
M	PRCC (A24) $\times$ CSHO (A25)	Market value: Price Close - Annual $\times$ Common Shares Outstanding measured at the end of the first quarter after year-end.
B	CEQ (A60)	Book value of of equity: Common/Ordinary Equity
B_{t-1}		Lagged book value
E	NI (A172)	Earnings, Net Income.
D	DVC (A21)	Dividends Common or Ordinary.
C	DVP (A19)	Preferred/Preference Dividends, or Coupons
O		Other items: $\Delta B - E + D + C$
z		A scalar, e.g. a foreign exchange rate
κ		Intercept term in the single-variable log-linear model
v		Intercept in the multi-variable, multiplicative (log-linear) model
β		Response coefficient (elasticity) in the single-variable log-linear model
β_i		Response coefficients (elasticities) in the multi-variable multiplica- tive (log-linear) model
k		Intercept term in the single- variable additive-linear model
v		Intercept in the multi-variable additive-linear model
b		Response coefficient in the single-variable additive-linear model
b_i		Response coefficients in the multi-variable additive-linear model
n or Obs		Number of observations
r_i		Ratio of accounting variable A_i to B_t
μ		Geometric average of market to book ratio, $\frac{M_t}{B_t}$
μ'		Estimate of market to book ratio based on single-variable log-linear model, using geometric means
μ''		Estimate of market to book ratio based on multi-variable log-linear model, using geometric means
m		Arithmetic average of market to book ratio, $\frac{M_t}{B_t}$
m'		Estimate of market to book ratio based on single-variable additive- linear model using arithmetic means
m''		Estimate of market to book ratio based on multi-variable additive- linear model using arithmetic means

TABLE 2
Summary statistics

Variable	Unlogged data					Logged data				
	Obs	Mean	Std. Err.	Skewness	Kurtosis	Obs	Mean	Std.Err.	Skewness	Kurtosis
M	264,963	2,926.54	62.57	68.22	6,853.73	264,954	4.85	0.00	0.13	-0.02
B	264,963	926.40	13.67	31.07	1,711.11	264,963	4.19	0.00	-0.09	0.14
B^+	244,180	1,032.02	14.63	31.41	1,658.55	244,180	4.37	0.00	-0.02	0.15
B^-	20,783	-314.52	27.32	-26.23	759.88	20,783	2.02	0.02	0.30	0.15
E	264,963	116.34	2.74	30.70	2,329.03	264,963	2.41	0.00	0.05	0.18
E^+	173,289	217.71	3.93	34.13	1,746.33	173,289	2.65	0.01	0.02	0.16
E^-	91,674	-75.27	2.62	-70.16	7,063.58	91,674	1.95	0.01	0.01	0.21
D	264,963	50.09	0.78	23.00	846.76	107,825	2.30	0.01	0.01	0.14
D^+	107,825	123.08	1.90	14.83	351.95	107,825	2.30	0.01	0.01	0.14
C	264,963	3.94	0.46	240.79	76,549.59	43,712	0.08	0.01	-0.13	-0.26
C^+	43,712	23.91	2.76	97.91	12,650.03	43,712	0.08	0.01	-0.13	-0.26
O	264,963	1.10	2.17	3.64	2,910.52	262,274	-0.25	0.02	-3.58	13.19
O^+	180,815	87.83	2.22	55.15	5,382.99	180,815	0.13	0.02	-4.09	18.40
O^-	81,459	-191.38	5.02	-29.03	1,268.08	81,459	-1.09	0.03	-2.81	7.21
$m, (M/B_0)$	264,963	35.71	30.13	466.46	228,470.12	264,954	0.54	0.00	-0.36	5.27
$r_0, (B_t/B_0)$	264,963	-0.16	1.70	-273.64	103,614.25	264,963	0.08	0.00	-0.05	18.01
$r_1, (E/B_0)$	264,963	-0.07	0.12	-184.64	65,409.66	264,963	-0.91	0.00	0.69	0.67
$r_2, (D/B_0)$	264,963	0.11	0.02	240.28	72,689.58	107,825	-3.07	0.00	0.70	8.90
$r_3, (C/B_0)$	264,963	0.02	0.03	431.43	217,489.13	43,712	-3.89	0.01	0.93	1.46
$r_4, (O/B_0)$	264,963	-0.97	1.72	-264.66	99,828.90	262,274	-0.68	0.02	0.23	12.77

The table shows pooled data average values of variables extracted from Compustat database 1972-2025, 265,203 firm-year observations. Superscripts on the variables indicate whether they are positive (+) or negative (-). M is market value, B is book value of equity, E is earnings, D is dividends, C is preferred dividend, and O is other items (owners' transactions). m is the market to book ratio; r_0 is the growth rate of book value, r_1 is the ratio of earnings to book value, r_2 is the ratio of dividends to book value, r_3 is the ratio of preferred dividend to book value, and r_4 is the ratio of other items to book value. When the data is unlogged the averages are arithmetic. When the data is logged, the averages are geometric (signed, for ratios and variables that can be negative).

TABLE 3
Testing Hypotheses 1 and 2

Period	$M = \kappa B ^\beta$ $\ln(\kappa)$	$\frac{M}{ B } = \kappa' B ^{\beta'}$ $\ln(\kappa')$	$M = \kappa B ^\beta$ β	$\frac{M}{ B } = \kappa' B ^{\beta'}$ β'	$M = k + bB$ k	$\frac{M}{B} = k' + b'\frac{1}{B}$ b'	$M = k + bB$ b	$\frac{M}{B} = k' + b'\frac{1}{B}$ k'	Obs Obs
1972	0.27	0.27	1.00	-0.00	-35.61	1.18	2.08	1.69	2,443
1973	-0.39	-0.39	1.07	0.07	-13.31	1.69	1.47	1.11	2,463
1974	-0.69	-0.69	1.08	0.08	-12.13	1.32	1.18	0.84	2,461
1975	-0.33	-0.33	1.05	0.05	-51.55	2.96	1.59	0.94	2,320
1976	-0.30	-0.30	1.05	0.05	-43.79	2.34	1.43	0.96	2,578
1977	-0.01	-0.01	0.99	-0.01	-21.13	3.00	1.17	0.99	2,569
1978	0.18	0.18	0.96	-0.04	-33.94	17.79	1.26	0.58	2,550
1979	0.17	0.17	0.95	-0.05	-26.36	28.18	1.15	0.04	2,476
1980	0.50	0.50	0.93	-0.07	15.46	4.56	1.26	1.69	2,449
1981	0.48	0.48	0.89	-0.11	20.39	0.90	0.96	1.94	4,006
1982	0.89	0.89	0.86	-0.14	57.90	7.07	1.11	1.09	4,650
1983	0.96	0.96	0.86	-0.14	110.85	4.00	0.86	1.99	4,904
1984	0.92	0.92	0.87	-0.13	5.69	14.63	1.43	-0.47	4,934
1985	1.06	1.06	0.88	-0.12	47.07	6.95	1.62	1.56	4,861
1986	1.24	1.24	0.86	-0.14	60.01	55.25	1.78	-13.75	5,100
1987	1.03	1.03	0.86	-0.14	75.84	8.59	1.45	1.03	5,224
1988	1.01	1.01	0.87	-0.13	91.16	5.05	1.50	2.29	5,102
1989	0.98	0.98	0.87	-0.13	83.04	3.51	1.66	3.56	4,986
1990	0.81	0.81	0.89	-0.11	78.53	7.39	1.77	1.58	4,995
1991	1.16	1.16	0.86	-0.14	116.24	4.86	1.84	3.81	5,122
1992	1.15	1.15	0.89	-0.11	46.66	31.09	2.36	-0.21	5,300
1993	1.29	1.29	0.87	-0.13	40.03	4.92	2.31	4.14	5,697
1994	1.09	1.09	0.90	-0.10	9.75	0.53	2.42	4.70	6,497
1995	1.27	1.27	0.88	-0.12	13.72	21.77	2.82	-0.36	7,165
1996	1.11	1.11	0.92	-0.08	-30.15	4.85	3.06	3.85	7,652
1997	1.22	1.22	0.94	-0.06	-94.28	11.97	4.16	3.13	7,687
1998	1.01	1.01	0.92	-0.08	-154.98	14.31	4.37	4.00	7,413
1999	1.59	1.59	0.83	-0.17	57.10	129.83	4.40	-29.53	7,431
2000	0.89	0.89	0.90	-0.10	58.56	14.55	3.26	5.51	7,423
2001	0.93	0.93	0.90	-0.10	309.39	2.47	2.53	8.76	6,978
2002	0.83	0.83	0.89	-0.11	16.18	2.59	2.45	6.34	6,649
2003	1.59	1.59	0.84	-0.16	53.59	9.04	2.85	7.11	6,288
2004	1.51	1.51	0.86	-0.14	225.17	10.22	2.48	6.69	6,188
2005	1.58	1.58	0.87	-0.13	406.33	7.91	2.42	7.69	5,922
2006	1.51	1.51	0.88	-0.12	511.63	15.63	2.31	5.85	5,755
2007	1.19	1.19	0.89	-0.11	463.48	2.74	2.04	8.18	5,541
2008	0.70	0.70	0.87	-0.13	452.10	0.89	1.33	7.11	5,230
2009	1.14	1.14	0.89	-0.11	857.22	7.70	1.53	13.67	5,032
2010	1.32	1.32	0.88	-0.12	1,173.64	4.80	1.52	8.20	4,825
2011	1.39	1.39	0.86	-0.14	1,290.07	27.48	1.41	1.53	4,787
2012	1.47	1.47	0.86	-0.14	1,414.15	5.41	1.55	8.53	4,687
2013	1.74	1.74	0.85	-0.15	1,683.55	97.60	1.70	-15.28	4,851
2014	1.71	1.71	0.85	-0.15	1,977.92	66.06	1.70	-10.56	4,943
2015	1.40	1.40	0.87	-0.13	1,907.41	548.20	1.63	-188.03	4,798
2016	1.54	1.54	0.88	-0.12	2,186.56	1.70	1.87	14.44	4,609
2017	1.70	1.70	0.86	-0.14	2,448.81	79.30	1.98	-0.12	4,482
2018	1.58	1.58	0.87	-0.13	2,653.17	51.13	2.00	434.45	4,415
2019	1.28	1.28	0.86	-0.14	2,419.95	15.38	1.78	41.03	4,317
2020	2.12	2.12	0.83	-0.17	3,668.36	6.22	2.70	22.27	4,387
2021	1.51	1.51	0.90	-0.10	2,933.41	16.58	3.03	20.91	4,593
2022	1.09	1.09	0.93	-0.07	2,544.22	10.05	2.60	9.15	4,590
2023	1.10	1.10	0.94	-0.06	2,532.38	74.98	3.34	24.06	4,452
2024	1.00	1.00	0.95	-0.05	2,335.18	5.96	3.70	8.30	4,299
2025	1.21	1.21	0.94	-0.06	1,227.60	80.52	4.44	27.38	3,878
Period	$\ln(\kappa)$	$\ln(\kappa')$	β	β'	k	b'	b	k'	Obs
Pooled	1.05	1.05	0.91	-0.09	430.05	65.54	2.69	-5.24	264,954
Average	1.01	1.01	0.90	-0.10	706.71	28.99	2.12	9.01	
Std	0.59	0.59	0.06	0.06	1,026.85	77.18	0.91	65.35	
SE	0.08	0.08	0.01	0.01	139.74	10.50	0.12	8.89	
LCL	0.85	0.85	0.89	-0.11	426.43	7.92	1.88	-8.83	
UCL	1.17	1.17	0.92	-0.08	986.98	50.06	2.37	26.84	
ρ		1.00		1.00		0.26		0.04	

The table shows comparison of single-variable models. Models of market value on book value are compared to algebraically equivalent single-variable models in which the dependent variable is the market to book ratio. If the market value models have the correct form, the estimates in the second pair of columns should be such that $\beta' = \beta - 1$. The parameter estimates in the other column pairs should be close to being equal. Estimates are based on Compustat data from 1972-2025. All models are estimated using OLS (per-year closed-form bivariate regression, i.e. Fama-MacBeth by fiscal year). Variables are defined in Table 1.

TABLE 4
Values of the Lilliefors test statistic for normality.

Variable	Obs.	Lilliefors statistic Unlogged data	Obs.	Lilliefors statistic Logged absolute data
<i>M</i>	264,963	0.46	264,954	0.02
<i>B</i>	264,963	0.43	264,963	0.00
<i>B</i> ⁺	244,180	0.44	244,180	0.01
<i>B</i> ⁻	20,783	0.47	20,783	0.05
<i>E</i>	264,963	0.40	264,736	0.01
<i>E</i> ⁺	173,130	0.45	173,130	0.01
<i>E</i> ⁻	91,606	0.46	91,606	0.01
<i>D</i>	264,963	0.45	107,572	0.02
<i>D</i> ⁺	107,572	0.42	107,572	0.02
<i>C</i>	264,963	0.49	43,126	0.03
<i>C</i> ⁺	43,126	0.48	43,126	0.03
<i>O</i>	244,570	0.41	244,570	0.01
<i>O</i> ⁺	171,454	0.46	171,454	0.01
<i>O</i> ⁻	73,116	0.44	73,116	0.02

The tables shows closeness of fit results for our main variables using the Lilliefors statistic. This statistic is interpreted as the maximum distance that the empirical distribution is from, in this case, the normal distribution. The Lilliefors statistic is based on the Kolmogorov-Smirnov test for goodness of fit. Unlike the latter, the former does not require knowledge of the population mean or standard deviation. *M* is market value, *B* is book value of equity, *E* is earnings, *D* is dividends, *C* is preference dividends, and *O* is other items. Superscripts on the variables indicate whether they are positive (+), negative (-). Variables are defined in [Table 1](#). Small amounts, i.e. below \$5,000 for *E*, *D*, *C*, and *O* are set to zero.

TABLE 5
Parameter Estimates for the models: $M = \kappa|B|^\beta$ and $M = v \prod_i |A_i|^{\beta_i}$

Period	κ	β_{B_t}	$\bar{R}^2$	v	$\beta_{B_{t-1}}$	β_E	β_D	β_C	β_O	$\Sigma_i \beta_i$	$\bar{R}^2$	Obs
1972	1.31	1.00	0.82	5.50	0.43	0.45	0.14	-0.01	0.01	1.01	0.85	2,443
1973	0.68	1.07	0.85	3.10	0.46	0.48	0.14	-0.01	0.01	1.07	0.88	2,463
1974	0.50	1.08	0.86	1.84	0.56	0.37	0.16	-0.02	0.00	1.08	0.87	2,461
1975	0.72	1.05	0.87	2.35	0.60	0.33	0.14	-0.01	0.01	1.07	0.88	2,320
1976	0.74	1.05	0.89	2.55	0.57	0.35	0.15	-0.02	0.01	1.06	0.91	2,578
1977	0.99	0.99	0.89	3.48	0.49	0.37	0.13	-0.00	0.01	1.00	0.91	2,569
1978	1.20	0.96	0.87	4.25	0.42	0.42	0.13	-0.01	0.01	0.98	0.90	2,550
1979	1.18	0.95	0.84	4.38	0.38	0.44	0.12	-0.02	0.01	0.94	0.87	2,476
1980	1.64	0.93	0.83	5.88	0.40	0.43	0.10	-0.02	0.01	0.92	0.84	2,449
1981	1.62	0.89	0.83	5.15	0.41	0.36	0.14	0.03	0.02	0.95	0.82	4,006
1982	2.44	0.86	0.78	5.68	0.50	0.30	0.10	-0.01	0.02	0.91	0.79	4,650
1983	2.61	0.86	0.82	8.25	0.41	0.33	0.14	-0.02	0.02	0.88	0.79	4,904
1984	2.51	0.87	0.82	5.69	0.50	0.29	0.15	-0.03	0.02	0.92	0.81	4,934
1985	2.89	0.88	0.81	6.11	0.54	0.25	0.16	-0.02	0.02	0.95	0.79	4,861
1986	3.46	0.86	0.80	9.09	0.44	0.29	0.21	-0.02	0.03	0.94	0.78	5,100
1987	2.81	0.86	0.81	6.56	0.47	0.28	0.20	-0.04	0.02	0.94	0.80	5,224
1988	2.75	0.87	0.80	5.43	0.52	0.27	0.18	-0.04	0.02	0.95	0.80	5,102
1989	2.67	0.87	0.78	5.36	0.51	0.28	0.20	-0.06	0.02	0.95	0.79	4,986
1990	2.26	0.89	0.77	4.82	0.52	0.24	0.26	-0.06	0.02	0.99	0.77	4,995
1991	3.18	0.86	0.75	8.94	0.42	0.28	0.27	-0.04	0.04	0.96	0.75	5,122
1992	3.17	0.89	0.78	10.70	0.40	0.30	0.24	0.01	0.04	1.00	0.78	5,300
1993	3.62	0.87	0.79	12.18	0.38	0.33	0.21	0.00	0.03	0.96	0.78	5,697
1994	2.98	0.90	0.80	8.10	0.46	0.34	0.15	-0.01	0.03	0.98	0.79	6,497
1995	3.58	0.88	0.76	10.00	0.44	0.36	0.14	-0.00	0.04	0.97	0.76	7,165
1996	3.03	0.92	0.79	9.09	0.46	0.36	0.13	-0.01	0.04	0.98	0.76	7,652
1997	3.38	0.94	0.79	8.03	0.55	0.31	0.12	-0.01	0.05	1.03	0.77	7,687
1998	2.73	0.92	0.74	5.91	0.53	0.32	0.16	-0.02	0.05	1.05	0.74	7,413
1999	4.90	0.83	0.63	14.17	0.26	0.52	0.09	0.05	0.06	0.98	0.65	7,431
2000	2.43	0.90	0.71	5.21	0.52	0.27	0.22	0.01	0.06	1.08	0.69	7,423
2001	2.53	0.90	0.73	4.24	0.65	0.12	0.28	-0.04	0.06	1.07	0.72	6,978
2002	2.29	0.89	0.73	3.94	0.61	0.14	0.27	-0.02	0.06	1.06	0.72	6,649
2003	4.89	0.84	0.75	9.21	0.49	0.32	0.15	0.01	0.05	1.01	0.77	6,288
2004	4.54	0.86	0.78	8.63	0.49	0.34	0.13	-0.03	0.05	0.98	0.80	6,188
2005	4.85	0.87	0.78	8.99	0.51	0.32	0.08	0.03	0.06	1.01	0.80	5,922
2006	4.51	0.88	0.78	8.36	0.52	0.31	0.09	0.03	0.07	1.03	0.80	5,755
2007	3.27	0.89	0.78	5.91	0.54	0.29	0.11	-0.01	0.07	1.00	0.78	5,541
2008	2.02	0.87	0.71	3.04	0.61	0.14	0.19	-0.06	0.06	0.94	0.70	5,230
2009	3.12	0.89	0.76	5.20	0.57	0.24	0.18	-0.05	0.09	1.02	0.79	5,032
2010	3.75	0.88	0.77	7.28	0.49	0.35	0.15	-0.04	0.06	1.01	0.80	4,825
2011	4.02	0.86	0.75	7.44	0.45	0.38	0.15	-0.01	0.05	1.02	0.79	4,787
2012	4.37	0.86	0.77	7.59	0.48	0.34	0.14	-0.00	0.06	1.02	0.81	4,687
2013	5.72	0.85	0.78	10.28	0.44	0.41	0.10	0.01	0.05	1.00	0.81	4,851
2014	5.54	0.85	0.76	8.28	0.49	0.36	0.11	-0.02	0.06	0.99	0.80	4,943
2015	4.06	0.87	0.75	6.52	0.53	0.27	0.15	-0.08	0.06	0.94	0.78	4,798
2016	4.66	0.88	0.76	7.71	0.52	0.31	0.16	-0.05	0.05	0.99	0.79	4,609
2017	5.49	0.86	0.76	8.82	0.50	0.35	0.12	-0.07	0.04	0.94	0.80	4,482
2018	4.88	0.87	0.75	8.47	0.47	0.36	0.13	-0.08	0.05	0.94	0.77	4,415
2019	3.60	0.86	0.70	5.70	0.49	0.34	0.13	-0.11	0.05	0.90	0.72	4,317
2020	8.32	0.83	0.71	15.02	0.41	0.37	0.10	-0.09	0.08	0.88	0.75	4,387
2021	4.54	0.90	0.75	8.69	0.49	0.38	0.10	-0.08	0.04	0.93	0.77	4,593
2022	2.97	0.93	0.76	3.85	0.60	0.29	0.16	-0.10	0.03	0.99	0.76	4,590
2023	3.01	0.94	0.77	3.76	0.62	0.31	0.12	-0.10	0.05	1.00	0.77	4,452
2024	2.72	0.95	0.77	3.72	0.62	0.28	0.14	-0.11	0.06	1.00	0.78	4,299
2025	3.35	0.94	0.76	4.79	0.59	0.31	0.10	-0.09	0.07	0.98	0.78	3,878
Period	κ	β_{B_t}	$\bar{R}^2$	v	$\beta_{B_{t-1}}$	β_E	β_D	β_C	β_O	$\Sigma_i \beta_i$	$\bar{R}^2$	Obs
Pooled	2.85	0.91	0.77	6.60	0.50	0.33	0.14	-0.02	0.04	0.99	0.78	264,954
Average	3.17	0.90	0.78	6.73	0.50	0.32	0.15	-0.03	0.04	0.98	0.79	
Std	1.50	0.06	0.05	2.83	0.07	0.07	0.05	0.04	0.02	0.05	0.05	
SE	0.20	0.01	0.01	0.38	0.01	0.01	0.01	0.01	0.00	0.01	0.01	
LCL	2.76	0.89	0.77	5.96	0.48	0.30	0.14	-0.04	0.03	0.97	0.78	
UCL	3.58	0.92	0.79	7.50	0.52	0.34	0.17	-0.02	0.05	1.00	0.80	

The table shows coefficient estimates of the models below, using data from 1972–2025, where the coefficients represent elasticities on variables $\{A_0..A_4\} = \{B_{t-1}, E_t, D_t, C_t, O_t\}$:

$$\begin{aligned}\ln(M) &= \ln(\kappa) + \beta \ln|B| & + & \varepsilon \\ \ln(M) &= \ln(v) + \sum_i \beta_i \ln|A_i| & + & \varepsilon,\end{aligned}$$

M is market value, B is book value of equity, E is earnings, D is dividends, C is preferred dividend, and O is other items. $\Sigma_i \beta_i$ is the sum of β_i . v is the exponentiated intercept. All models are estimated by taking logs and using OLS.

TABLE 6
Decomposition of the market to book ratio

Period	μ	μ'	μ''	m	m'	m''	Obs
1972	1.30	1.30	1.27	1.54	1.87	3.39	2,443
1973	0.89	0.89	0.88	1.17	1.40	0.86	2,463
1974	0.69	0.69	0.69	0.90	1.12	0.81	2,461
1975	0.89	0.89	0.87	1.11	1.31	0.22	2,320
1976	0.91	0.91	0.90	1.06	1.24	0.82	2,578
1977	0.95	0.95	0.93	1.10	1.09	0.94	2,569
1978	1.02	1.02	1.01	1.49	1.14	1.26	2,550
1979	0.94	0.94	0.94	1.35	1.07	1.15	2,476
1980	1.22	1.22	1.22	1.76	1.31	1.31	2,449
1981	1.09	1.09	0.96	1.56	1.04	0.99	4,006
1982	1.56	1.56	1.47	2.19	1.35	1.50	4,650
1983	1.65	1.65	1.55	2.43	1.31	0.43	4,904
1984	1.65	1.65	1.57	3.40	1.45	0.07	4,934
1985	1.93	1.93	1.80	3.18	1.80	0.79	4,861
1986	2.17	2.17	1.94	0.36	2.01	1.57	5,100
1987	1.79	1.79	1.68	3.09	1.73	1.00	5,224
1988	1.78	1.78	1.69	3.24	1.82	1.02	5,102
1989	1.73	1.73	1.68	2.64	1.94	1.80	4,986
1990	1.54	1.54	1.46	2.78	2.02	1.47	4,995
1991	1.92	1.92	1.82	2.72	2.20	1.08	5,122
1992	2.09	2.09	1.98	3.76	2.51	4.12	5,300
1993	2.25	2.25	2.17	4.09	2.43	2.23	5,697
1994	2.03	2.03	1.99	3.64	2.45	2.18	6,497
1995	2.33	2.33	2.26	4.51	2.86	2.28	7,165
1996	2.25	2.25	2.22	3.49	2.97	-0.56	7,652
1997	2.68	2.68	2.64	3.65	3.91	-0.47	7,687
1998	2.02	2.02	2.00	6.20	4.00	3.94	7,413
1999	2.53	2.53	2.51	19.53	4.52	1.35	7,431
2000	1.61	1.61	1.63	7.77	3.37	4.48	7,423
2001	1.75	1.75	1.74	1.22	3.05	5.07	6,978
2002	1.47	1.47	1.44	3.56	2.48	0.88	6,649
2003	2.62	2.62	2.63	5.42	2.92	3.05	6,288
2004	2.56	2.56	2.59	0.83	2.75	3.42	6,188
2005	2.73	2.73	2.80	4.65	2.87	3.37	5,922
2006	2.64	2.64	2.75	3.36	2.80	5.51	5,755
2007	2.00	2.00	2.10	6.61	2.44	3.20	5,541
2008	1.13	1.13	1.27	1.42	1.78	2.16	5,230
2009	1.84	1.84	1.94	14.30	2.24	10.36	5,032
2010	2.15	2.15	2.27	3.43	2.37	1.78	4,825
2011	2.04	2.04	2.18	-2.03	2.29	0.30	4,787
2012	2.25	2.25	2.41	3.91	2.44	5.35	4,687
2013	2.77	2.77	2.90	10.97	2.71	6.13	4,851
2014	2.68	2.68	2.79	7.62	2.86	-0.83	4,943
2015	2.17	2.17	2.32	233.52	2.74	5.39	4,798
2016	2.61	2.61	2.79	6.02	3.07	4.24	4,609
2017	2.71	2.71	2.80	20.76	3.20	-3.43	4,482
2018	2.51	2.51	2.59	463.43	3.27	-66.65	4,415
2019	1.75	1.75	1.78	39.97	2.86	-9.22	4,317
2020	3.36	3.36	3.39	-5.09	4.28	21.10	4,387
2021	2.63	2.63	2.72	22.68	4.21	-14.39	4,593
2022	2.04	2.04	2.20	3.49	3.62	-0.53	4,590
2023	2.20	2.20	2.25	27.46	4.27	-9.37	4,452
2024	2.07	2.07	2.11	3.71	4.47	-8.30	4,299
2025	2.32	2.32	2.29	-26.76	4.79	59.59	3,878
Period	μ	μ'	μ''	m	m'	m''	Obs
Pooled	1.95	1.95	1.98	17.02	3.16	-0.65	264,954
Average	1.93	1.93	1.94	17.60	2.52	1.30	
Std	0.61	0.61	0.64	69.75	1.01	13.20	
SE	0.08	0.08	0.09	9.49	0.14	1.80	
LCL	1.77	1.77	1.76	-1.44	2.24	-2.30	
UCL	2.10	2.10	2.12	36.63	2.79	4.90	

The table shows estimates of the Market to Book ratio based on the following models, estimated using Compustat data from 1972-2025.

$$\begin{aligned}
 \mu &= \frac{M_t}{|B_t|} & m &= \frac{M}{B} \\
 \mu' &= \kappa |B_t|^{\beta-1} & m' &= \frac{k}{B} + b \\
 \mu'' &= v \left(\prod_i |r_i|^{\beta_i} \right) |B_t|^{\sum \beta_i - 1} & m'' &= \frac{v}{B_t} + \sum_i b_i r_i.
 \end{aligned}$$

The estimates are based on geometric (arithmetic) averages of the variables in the case of the log-linear (additive-linear) models.

TABLE 7
Parameter Estimates form partitioned model

Partition	B_{t-1}	E	D	C	O	ν	$\beta_{B_{t-1}}$	β_E	β_D	β_C	β_O	$\Sigma_i \beta_i$	$\bar{R}^2$	Obs
1	+	+	+	+	-	4.15	0.44	0.42	0.22	-0.03	0.01	1.05	0.94	6,567
2	+	+	+	+	0	2.40	0.40	0.67	0.12	-0.08	0.00	1.11	0.97	33
3	+	+	+	+	+	3.63	0.53	0.35	0.17	-0.06	0.03	1.02	0.89	14,177
4	+	+	+	0	-	7.80	0.33	0.51	0.22	0.00	0.01	1.07	0.94	34,374
5	+	+	+	0	0	7.65	0.24	0.59	0.17	0.00	0.00	1.01	0.88	466
6	+	+	+	0	+	9.00	0.37	0.48	0.15	0.00	0.02	1.02	0.90	41,473
7	+	+	0	+	-	4.09	0.55	0.41	0.00	0.05	0.01	1.02	0.84	2,061
8	+	+	0	+	0	5.19	0.25	0.50	0.00	-0.07	0.00	0.68	0.53	45
9	+	+	0	+	+	13.30	0.36	0.42	0.00	0.05	0.06	0.89	0.76	5,056
10	+	+	0	0	-	3.86	0.62	0.44	0.00	0.00	0.01	1.07	0.90	20,457
11	+	+	0	0	0	5.46	0.33	0.61	0.00	0.00	0.00	0.95	0.65	488
12	+	+	0	0	+	11.11	0.47	0.43	0.00	0.00	0.04	0.94	0.82	41,961
13	+	-	+	+	-	1.28	0.83	-0.08	0.27	-0.00	0.01	1.02	0.87	682
15	+	-	+	+	+	5.10	0.58	-0.04	0.30	0.00	0.07	0.91	0.77	1,411
16	+	-	+	0	-	1.36	0.86	-0.07	0.23	0.00	0.01	1.03	0.86	2,969
17	+	-	+	0	0	1.16	0.77	0.06	0.01	0.00	0.00	0.84	0.64	119
18	+	-	+	0	+	3.99	0.68	0.02	0.22	0.00	0.04	0.95	0.80	4,333
19	+	-	0	+	-	2.55	0.73	-0.07	0.00	0.15	0.02	0.82	0.62	1,225
20	+	-	0	+	0	2.29	0.53	-0.18	0.00	0.17	0.00	0.52	0.31	65
21	+	-	0	+	+	8.96	0.40	0.11	0.00	0.10	0.12	0.73	0.48	7,365
22	+	-	0	0	-	1.30	0.89	-0.02	0.00	0.00	0.01	0.88	0.72	10,452
23	+	-	0	0	0	1.89	0.38	0.10	0.00	0.00	0.00	0.47	0.27	1,146
24	+	-	0	0	+	6.96	0.52	0.19	0.00	0.00	0.07	0.78	0.60	46,187
B^-						8.34	0.20	0.45	0.44	0.04	0.07	1.20	0.52	21,835
Averages						5.12	0.51	0.26	0.10	0.01	0.03	0.92	0.73	264,947
Estimates weighted by observations						7.42	0.46	0.36	0.11	0.00	0.04	0.97	0.78	
Pooled data parameter estimates						6.60	0.50	0.33	0.14	-0.02	0.04	0.99	0.78	264,954

The table shows coefficient estimates of the following model: $M = \nu \prod_i |A_i|^{\beta_i}$, where the slope coefficients are estimated from the second model in (7), for Compustat data during 1972-2025. M is market value, B is book value of equity, E is earnings, D is dividends, C is preferred dividend, and O is other items. $\Sigma_i \beta_i$ is the sum of β_i . β_i denotes the elasticities on $\{A_0..A_4\} \in \{B_{t-1}, E_t, D_t, C_t, O_t\}$; ν is the exponentiated intercept of that model. The symbols in columns 2 through 6 indicate whether the variable in that column is positive (+), zero (0), or negative (-). Partitions with too few observations to estimate the model are omitted. All models are estimated by taking logs and using OLS.